Chapter 2 - Functions and Continuity

Calculus I

August 30, 2024

Table of Contents

1. The Notion of a Function
2. Limits of Functions
3. Arithmetic Rules for Limits of Functions
4. One-sided Limits and Limits at Infinity
5. Continuity
6. Continuity on an Interval

Section 1

The Notion of a Function

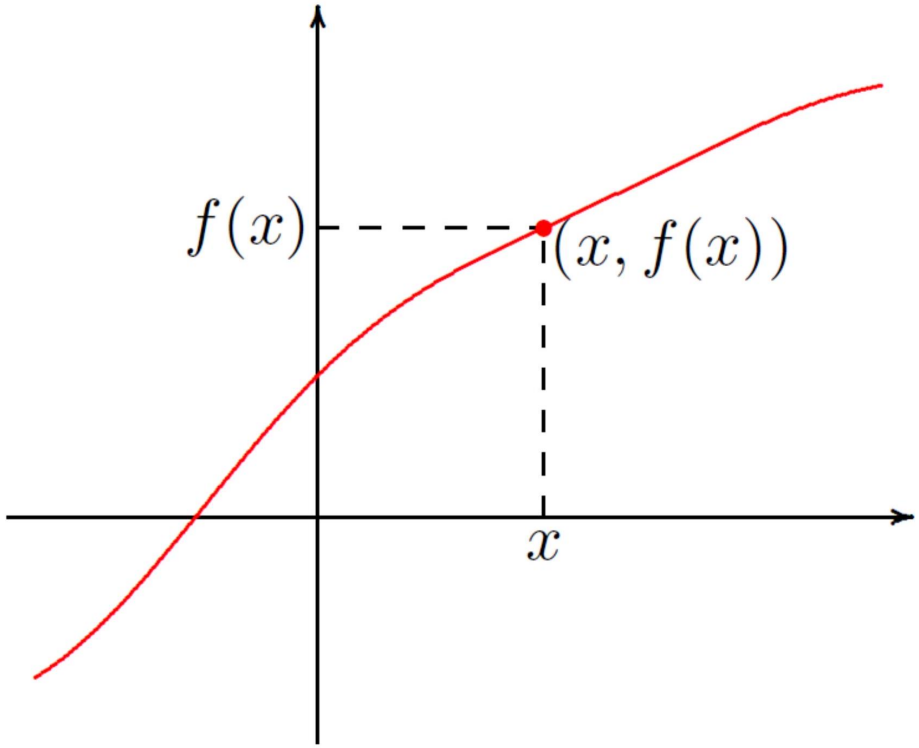
Functions

Definition 2.1.1

Let D, E be sets. A function $f : D \to E$ is a rule that assigns each element x in D exactly one element, named $f(x)$, in E. Here, $f(x)$ is called the value of f at x, D is called the domain of f and the set $\{f(x) \mid x \in D\}$ is called the range of f.

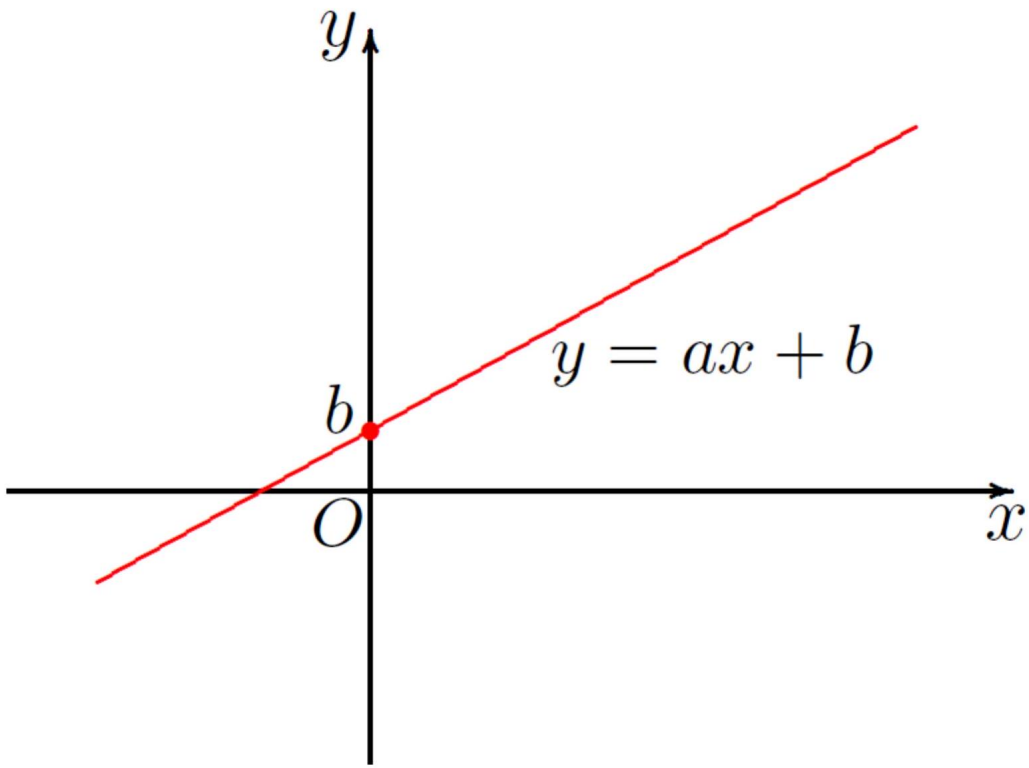
Graph of a function

Given a function $y = f(x)$. For each x, we get a point $(x, f(x))$ on the coordinate plane. When x moves in the domain D of f, the point $(x, f(x))$ moves on the plane. The trace of $(x, f(x))$ for $x \in D$ is called the graph of f.



Example

Consider the linear function $f(x) = ax + b$. We know that the graph of f is a straight line.



The coefficient a in $f(x) = ax + b$ is called the slope of the line and b is the intercept of the line since the intersection of the L with the y-axis is $(0, b)$.

Basic Functions

- Absolute values. The absolute value of a number a is denoted by $|a|$ and defined by

$$\begin{cases} |a| = a & \text{if } a > 0 \\ |a| = -a & \text{if } a \leq 0 \end{cases}$$

  One may regard $|\cdot|$ as a function with domain $\mathbb{R}$ and range $[0, \infty)$.

- Polynomials. A polynomial is a function P of the following form

$$P(x) = a_0 + \sum_{k=1}^{n} a_k x^k = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n, \quad \forall x \in \mathbb{R}$$

  where n is a nonnegative integer, which is called the degree of P when $a_n \neq 0$, and $a_0, \ldots, a_n$ are real numbers, which are called the coefficients of P. In the case that $n = 2$ and $a_2 \neq 0$, P is also called a quadratic function.

Basic Functions cont'd

- Power functions. A power function is a function of the form $f(x) = x^a$, where a is a constant. If $a \in \mathbb{N}$, then $f(x)$ is a polynomial with domain $\mathbb{R}$. If $a = 1/n$ with $n \in \mathbb{N}$, then f is a root function with domain $[0, \infty)$. If $a = -1$, then f is the reciprocal function with domain $(0, \infty)$.
- Rational functions. A rational function is a function of the form $f(x) = p(x)/q(x)$, where $p(x)$ and $q(x)$ are both polynomials. One can see that the domain of f is $\{x \in \mathbb{R} \mid q(x) \neq 0\}$.
- Algebraic functions. An algebraic function is a function constructed by using algebraic operators including addition, subtraction, multiplication, division and taking roots. For example, $\sqrt{x^2+1} - x - 2$ and $\left(x^{1/3} - 1\right) / \sqrt{x+4}$ are algebraic functions.

Basic Functions cont'd

- Trigonometric functions. Trigonometric functions consist of $\sin x, \cos x, \tan x, \cot x, \sec x$ and $\csc x$.
- Exponential functions. An exponential function is a function of the form $f(x) = b^x$, where $b > 0$ is a constant and called the base. The domain of f is $\mathbb{R}$ and its definition will be discussed in the next subsection.
- Logarithmic functions. A logarithmic function is a function of the form $f(x) = \log_b x$, where $b > 0$ is called the base. Logarithmic functions are defined to be the inverse function of exponential functions.

Combination of functions

Definition 2.1.2 (Combination of Functions)

Let f and g be functions with domains D_f, D_g.

1. The addition $f+g$ and multiplication fg are defined by

$$(f+g)(x) = f(x) + g(x), (fg)(x) = f(x)g(x) \quad \forall x \in D_f \cap D_g$$

2. The ratio f/g is a function with domain $\{x \in D_f \cap D_g \mid g(x) \neq 0\}$ and defined by

$$(f/g)(x) = f(x)/g(x)$$

3. The composition of f and g is denoted by $g \circ f$ and defined by

$$(g \circ f)(x) = g(f(x)), \quad \forall x \in D_f, f(x) \in D_g$$

Combination of functions cont'd

Example 2.1.3

Let $f(x) = \cos x$ and $g(x) = 2x$. Then,

$$(f \circ g)(x) = f(g(x)) = f(2x) = \cos(2x)$$

To avoid any confusion, we write $\cos(2x)$ for $f \circ g$ instead of $\cos 2x$.

Exponential Functions

Generally, an exponential function is a function of the form

$$f(x) = b^x$$

where b is a positive constant, called the base, and x is any real number.

- In the case that x is an integer, say $n, b^0 = 1$ if $n = 0$ and

$$b^n = \underbrace{b \times b \times \cdots \times b}_{n}, \quad b^{-n} = 1/b^n, \quad \forall n \in \mathbb{N}$$

- In the case that x is a rational number, say p/q with $p, q \in \mathbb{N}$ and $q \neq 0$,

$$b^{p/q} = \sqrt[q]{b^p} = (\sqrt[q]{b})^p$$

Exponential Functions cont'd

To see the meaning of b^x when x is irrational, we consider the case $x = \pi$.

Note that $\pi = 3.14159\cdots$. By the law of rational exponents, one can show that

$$2^3 < 2^{3.1} < 2^{3.14} < 2^{3.141} < 2^{3.1415} < \cdots$$

and

$$2^4 > 2^{3.2} > 2^{3.15} > 2^{3.142} > 2^{3.1416} > \cdots$$

Observe that

$$2^{3.2} - 2^{3.1} = 2^{3.1}\left(2^{0.1} - 1\right) < 16\left(2^{0.1} - 1\right)$$

Exponential Functions cont'd

Similarly, one can show that

$$2^{3.15} - 2^{3.14} < 16\left(2^{0.01} - 1\right)$$
$$2^{3.142} - 2^{3.141} < 16\left(2^{0.001} - 1\right)$$
$$2^{3.1416} - 2^{3.1415} < 16\left(2^{0.0001} - 1\right)$$

Inductively, we may approximate 2^{π} using the sequence

$$2^{3.1}, 2^{3.14}, 2^{3.141}, 2^{3.1415}, \ldots$$

or the sequence

$$2^{3.2}, 2^{3.15}, 2^{3.142}, 2^{3.1416}, \ldots$$

Law of exponents

Theorem 2.1.4

For $a, b \in (0, \infty)$ and $x, y \in \mathbb{R}$, one has

$$a^{x+y} = a^x a^y, a^{x-y} = a^x / a^y, (a^x)^y = (a^y)^x = a^{xy}, (ab)^x = a^x b^x$$

Remark

Let $f(x) = b^x$. Then, f is increasing if $b > 1$ and decreasing if $0 < b < 1$.

The number e

Consider the function $f(x) = b^x$ and let m be the slope of the line "tangent" to the curve $y = f(x)$ at $(0, 1)$. Then:

- $m > 0$ if $b > 1$;
- $m < 0$ if $0 < b < 1$;
- $m = 0$ if $b = 1$.

The slope m changes along with the base b and, in fact, the slope m, as a function of b, is increasing.

The number e can be defined in any of the following way.

- e is the base b such that the slope of the line tangent to the curve $y = b^x$ at $(0, 1)$ is 1.

Note that $e = 2.71828\cdots$.

Example

Consider the following function. Let $D = \{1, 2, 3\}, E = \{a, b, c\}$ and let $f : D \to E$ be a function defined by

$$f(1) = a, \quad f(2) = f(3) = b$$

- Given a, there is only one x such that $f(x) = a$.
- But, given b, the equation $f(x) = b$ has two solutions, which are 2 and 3 .

Injective Functions

Definition 2.1.5

A function f with domain D is one-to-one (briefly, 1-1) or injective if

$$\forall x, y \in D, x \neq y \quad \Rightarrow \quad f(x) \neq f(y)$$

Equivalently, $f(x) = f(y)$ implies $x = y$.

Remark

- A function is one-to-one if and only if no horizontal line intersects its graph more than once. This criterion is called the horizontal line test.
- Equivalently, f is one-to-one if $f(x) = c$ has at most one solution for all $c \in \mathbb{R}$.

Inverse Functions cont'd

Example 2.1.6

Let $f(x) = x^3$ and $g(x) = x^2$.

- Clearly, g is not 1-1 because $g(1) = g(-1) = 1$.
- For f, consider the follow computation,

$$x^3 - y^3 = (x - y)\left(x^2 + xy + y^2\right) = (x - y)\left[(x + y)^2 + x^2 + y^2\right]/2$$

If $x \neq y$, then $x^2 + y^2 > 0$, which implies $x^3 \neq y^3$. This proves that f is $1 - 1$.

Inverse Functions cont'd

Definition 2.1.7

Let f be a one-to-one function with domain D and range $\mathbb{R}$. The inverse function of f (denoted by f^{-1} and read as " f inverse") is defined by

$$f^{-1}(y) = x \quad \Leftrightarrow \quad f(x) = y$$

for all $y \in \mathbb{R}$.

Remark

We write the reciprocal of $f(x)$ as $[f(x)]^{-1}$ or $(1/f)(x)$.

Inverse Functions cont'd

- If f is $1-1$, then the domain of f^{-1} is the range of f and the range of f^{-1} is the domain of f. Furthermore, one has $\left(f^{-1}\right)^{-1} = f$.

Theorem 2.1.8 (Cancellation Equations)

If f is one-to-one with domain D and range R, then

$$\left(f^{-1} \circ f\right)(x) = x, \quad \forall x \in D, \quad \text{and} \quad \left(f \circ f^{-1}\right)(y) = y, \quad \forall y \in R$$

Example 2.1.9

Let $f(x) = x + 2$ and $g(x) = x^{1/3}$. Then,

- $f^{-1}(y) = y - 2$ and
- $g^{-1}(y) = y^3$.

Computing the inverse function

Write $y = f(x)$ and solve x in term of y, say $x = g(y)$. Then, g is the inverse of f.

Example 2.1.10

1. Let $f(x) = \left(x^3 - 2\right)^{1/5}$ for $x \in \mathbb{R}$. Write $y = f(x)$ and solve this equation to obtain $x = \left(y^5 + 2\right)^{1/3}$ for $y \in \mathbb{R}$. Then, the function $f^{-1}(y) = \left(y^5 + 2\right)^{1/3}$ is the inverse of f.
2. Let $f(x) = \sqrt{x} + 1$ for $x \in [0, \infty)$. Solving $y = f(x)$ yields $x = (y-1)^2$. This implies $f^{-1}(y) = (y-1)^2$ with $y \in [1, \infty)$ is the inverse of f.

Remark

The graph of f^{-1} is the reflection of the graph of f with respect to $y = x$.

Logarithmic functions

Let $b > 0$ and $b \neq 1$. Note that $b^x = b^y$ implies $b^{x-y} = 1$ and, hence, $x = y$.

- This proves that $f(x) = b^x$ is $1-1$ with domain $(-\infty, \infty)$ and range $(0, \infty)$.

The inverse function of f is called the logarithmic function with base b and denoted by $f^{-1}(y) = \log_b y$ with domain $(0, \infty)$ and range $(-\infty, \infty)$.

- By definition, one has $\log_b y = x$ if and only if $b^x = y$ for $y \in (0, \infty)$.
- This implies

$$b^{\log_b y} = y, \quad \forall y \in (0, \infty), \qquad \text{and} \qquad \log_b (b^x) = x, \quad \forall x \in \mathbb{R}$$

Law of logarithms

Theorem 2.1.11

For $b, x, y \in (0, \infty)$ and $r \in \mathbb{R}$, one has

$$\log_b(xy) = \log_b x + \log_b y$$
$$\log_b(x/y) = \log_b x - \log_b y$$
$$\log_b (x^r) = r \log_b x$$

Remark

When $b = e$, we also write $\ln x$ for $\log_e x$ and call it the natural logarithm.

Change of base formula

Theorem 2.1.12

Let a, b, x be positive constants and assume that $a, b \neq 1$. Then,

$$\log_b x = \frac{\log_a x}{\log_a b}$$

In particular, $\log_b x = \ln x / \ln b$.

Proof. It suffices to show the specific case, while the general case follows immediately. Set $y = \log_b x$. Then,

$$x = b^y = \left(e^{\ln b}\right)^y = e^{(\ln b)y}$$

This implies

$$(\ln b)y = \ln x$$

Inverse trigonometric functions

It is clear that none of the trigonometric functions is one-to-one on $\mathbb{R}$, unless it is restricted on a specific region.

- For example, $\sin x$ is one-to-one on $[-\pi/2, \pi/2]$.
  - We write $\sin^{-1}$ or arcsin for the inverse function, which has domain $[-1, 1]$ and range $[-\pi/2, \pi/2]$.
  - This means that, for $x \in [-1, 1]$ and $y \in [-\pi/2, \pi/2]$,

$$\sin^{-1} x = y \quad \Leftrightarrow \quad \sin y = x$$

Inverse trigonometric functions cont'd

Similarly, the inverse functions of the other trigonometric functions can be defined by

$$\cos^{-1} x = y \quad \Leftrightarrow \quad \cos y = x, \quad \forall x \in [-1, 1], y \in [0, \pi]$$
$$\tan^{-1} x = y \quad \Leftrightarrow \quad \tan y = x, \quad \forall x \in \mathbb{R}, y \in (-\pi/2, \pi/2)$$
$$\sec^{-1} x = y \quad \Leftrightarrow \quad \sec y = x, \quad \forall y \in [0, \pi/2) \cup (\pi/2, \pi], x \in (-\infty, -1] \cup [1, \infty)$$
$$\cot^{-1} x = y \quad \Leftrightarrow \cot y = x, \quad \forall y \in (0, \pi), x \in \mathbb{R}$$
$$\csc^{-1} x = y \quad \Leftrightarrow \quad \csc y = x, \quad \forall y \in [\pi/2, \pi) \cup (\pi, 3\pi/2], |x| \geq 1$$

Remark

Note that the range of $\sec^{-1}$ and $\csc^{-1}$ are not universally agreed on.

Inverse trigonometric functions cont'd

Example 2.1.13

Compute $\left(\cos \circ \sin^{-1}\right)\left(\frac{1}{\sqrt{2}}\right)$.

Solution. Note that $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = \pi/4$. This implies

$$\left(\cos \circ \sin^{-1}\right)\left(\frac{1}{\sqrt{2}}\right) = \cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

Inverse trigonometric functions cont'd

Example 2.1.14

Simplify $\sin\left(\tan^{-1} x\right)$.

Solution.

- We set $y = \tan^{-1} x$ or, equivalently, $\tan y = x$ with $y \in (-\pi/2, \pi/2)$.
- Suppose $x > 0$ and consider the right triangle with sides of lengths $1, x$ and $\sqrt{x^2+1}$. Observe that y is exactly the radian of the angle opposite to the side of length x. This gives

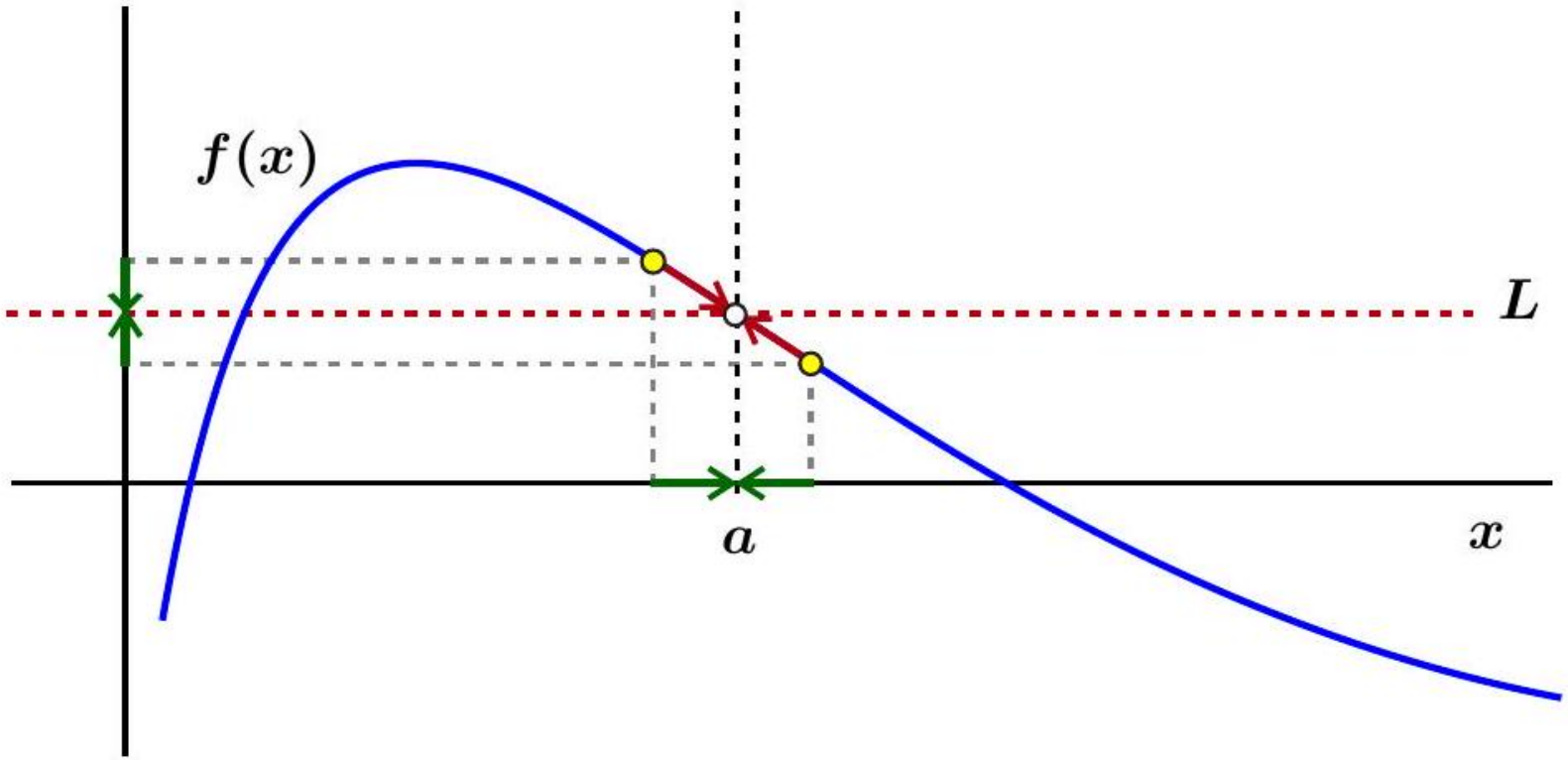
$$\sin\left(\tan^{-1} x\right) = \sin y = x/\sqrt{x^2+1}$$

- For $x < 0$, the discussion is similar and skipped.

Section 2

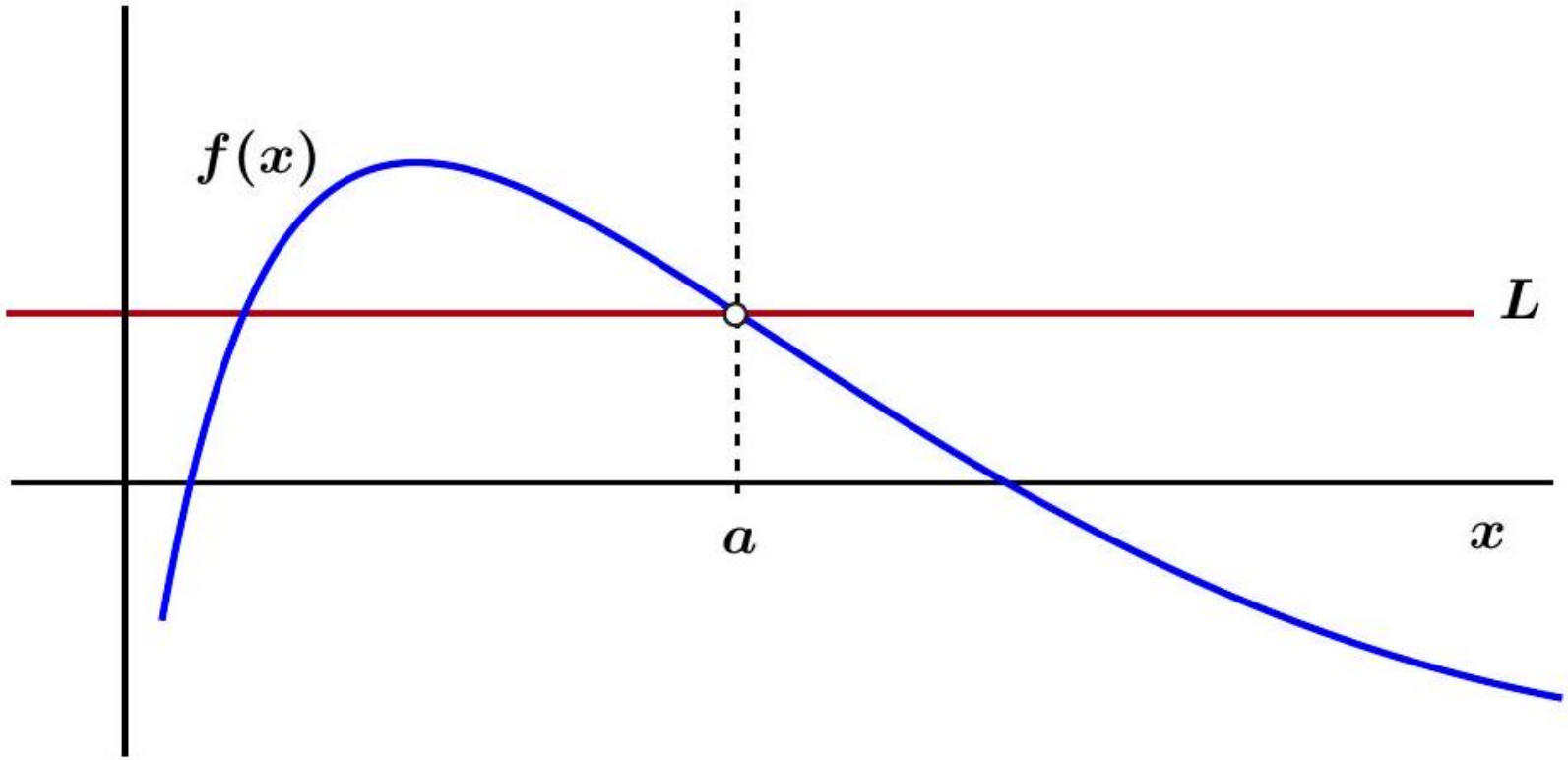
Limits of Functions

What is a Limit?



Heuristic Definition: We say that L is the limit of a function $f(x)$ as x approaches a if as x gets closer and closer to a, without ever reaching a, $f(x)$ gets closer and closer to L.

What is a Limit?



Heuristic Definition: We say that L is the limit of a function $f(x)$ as x approaches a if as x gets closer and closer to a, without ever reaching a, $f(x)$ gets closer and closer to L.

What is a Limit?

Example 2.2.1

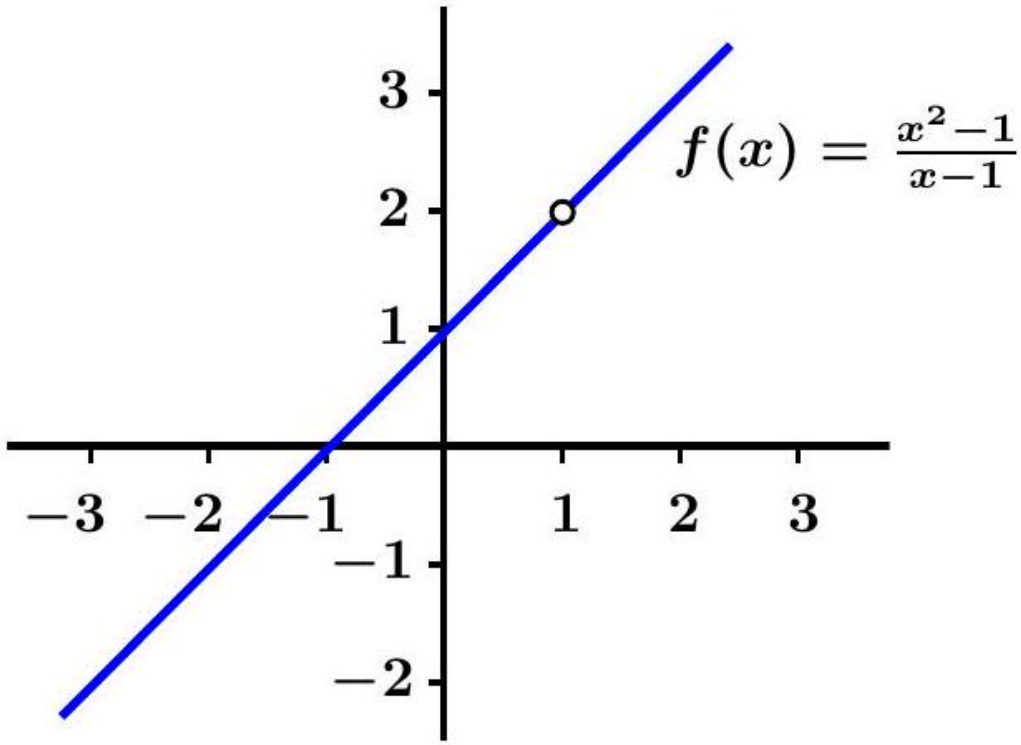
Consider the two functions $f(x) = \frac{x^2-1}{x-1}$ and $g(x) = x + 1$. It might be tempting to use some algebra and write

$$\begin{aligned} f(x) &= \frac{x^2 - 1}{x - 1} \\ &= \frac{(x+1)(x-1)}{x-1} \\ &= x + 1 \\ &= g(x) \end{aligned}$$

Question: Does this mean that $\boldsymbol{f(x)}$ and $\boldsymbol{g(x)}$ are actually the same function?

Answer: Almost, but not quite. They have different domains since $f(x)$ is not defined at $x = 1$!

What is a Limit?



Note: The graph of $g(x) = x + 1$ is a straight line with slope 1 .

Question: What happens if we graph $f(x) = \frac{x^2-1}{x-1}$?

Answer: The graph of $f(x)$ is the same graph as $g(x) = x + 1$, except there is a hole in the graph corresponding to where $x = 1$.

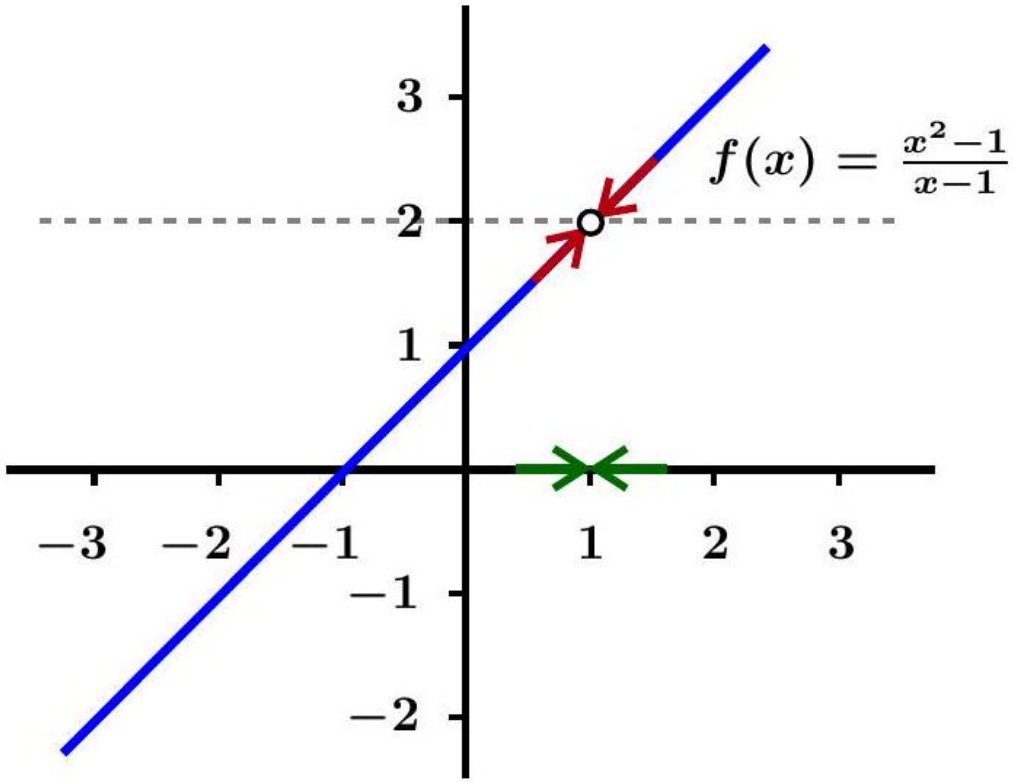
What is a Limit?

We want to focus on the values of $f(x) = \frac{x^2-1}{x-1}$ when x is very close to but not equal to 1. The following is a table of some select values with x near 1.

x	$f(x)$
0	1
0.1	1.1
0.5	1.5
0.75	1.75
0.9	1.9
0.99	1.99
0.999	1.999
0.99999	1.99999
0.99999999	1.99999999

x	$f(x)$
2	3
1.9	2.9
1.5	2.5
1.25	2.25
1.1	2.1
1.01	2.01
1.001	2.001
1.00001	2.00001
1.00000001	2.00000001

What is a Limit?



We can see that as x gets closer and closer to 1, $f(x)$ gets closer and closer to 2.

We would like to say that 2 is the limit of $\boldsymbol{f(x)}$ as $\boldsymbol{x}$ approaches 1.

Formal Definition of a Limit

A more robust definition is required.

Improved Heuristic Definition: $\boldsymbol{L}$ is the limit of $\boldsymbol{f(x)}$ as $\boldsymbol{x}$ approaches $\boldsymbol{a}$ if for any positive tolerance $\boldsymbol{\epsilon > 0}$, we can ensure that $\boldsymbol{f(x)}$ approximates L with error less than ϵ at any $\boldsymbol{x}$, other than possibly at $\boldsymbol{a}$ itself, provided that $\boldsymbol{x}$ is close enough to a.

Definition 2.2.2 (Limit of a Function)

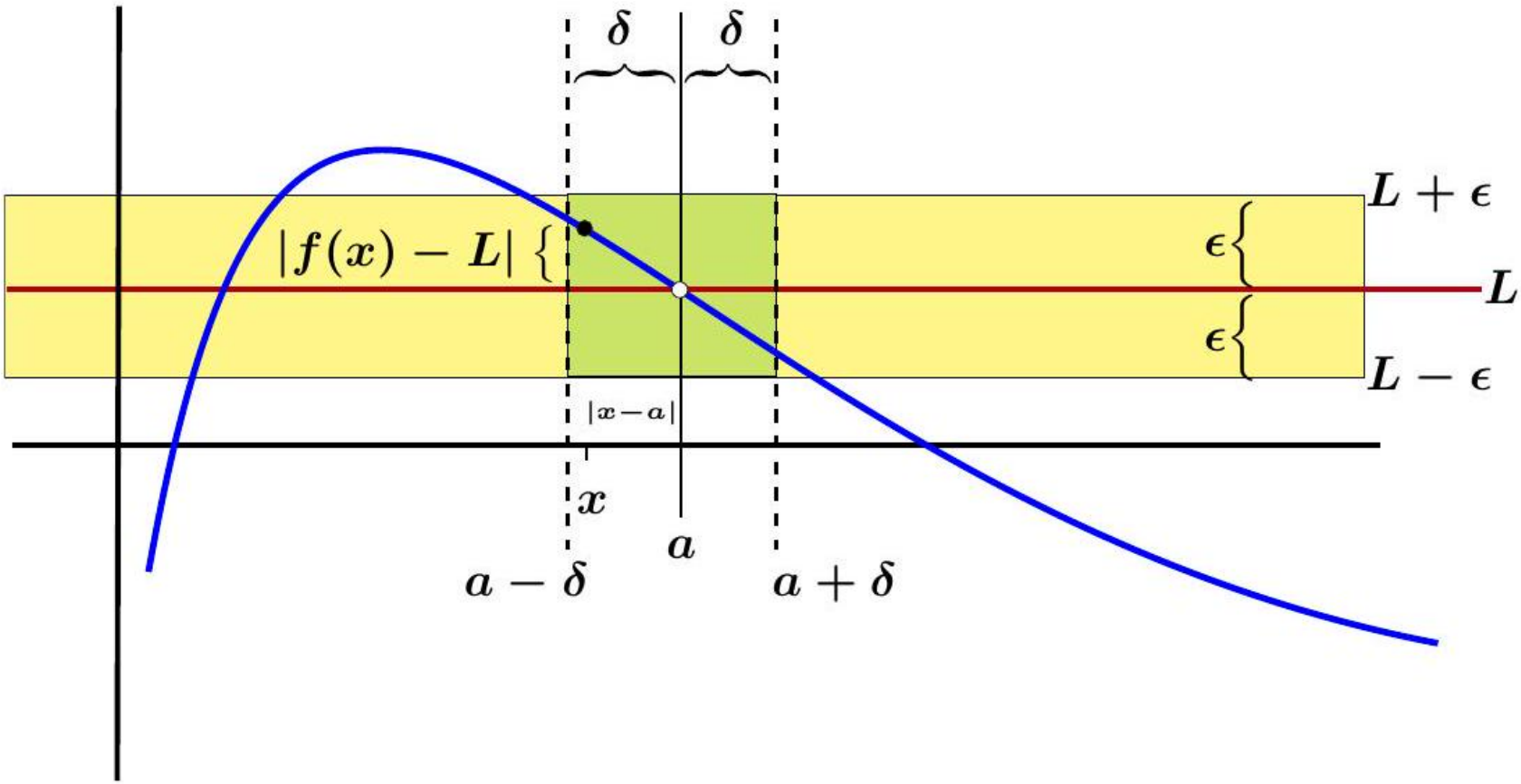
We say that $\boldsymbol{L}$ is the limit of $\boldsymbol{f(x)}$ as $\boldsymbol{x}$ approaches $\boldsymbol{a}$ if for every $\boldsymbol{\epsilon > 0}$ there exists a $\delta > 0$ such that if

$$0 < |x - a| < \delta$$
$$|f(x) - L| < \epsilon$$

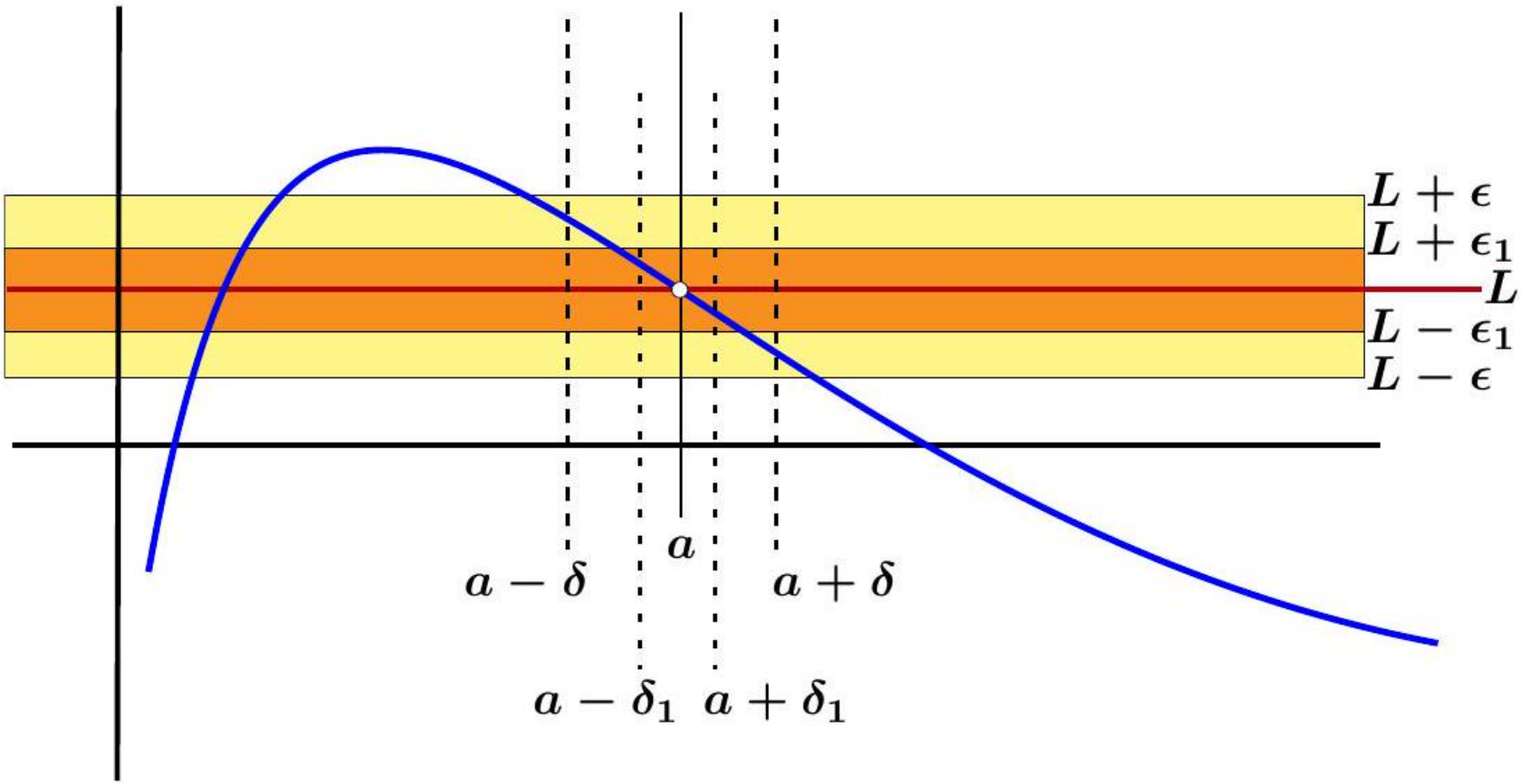
We write

$$\lim_{x \to a} f(x) = L$$

Formal Definition of a Limit



Formal Definition of a Limit



Remarks:

1. For $\lim_{x \to a} f(x)$ to exist, $f(x)$ must be defined on an open interval (α, β) containing $x = a$, except possibly at $x = a$.
2. The value of $f(\boldsymbol{a})$, if it is defined at all, does not affect the existence of the limit or its value.
3. If two functions are equal, except possibly at $x = a$, then their limiting behavior at $\boldsymbol{a}$ is the same.

Uniqueness of Limits

Exercise: Modify the argument in the previous example to prove:

Theorem 2.2.3 (Uniqueness of Limits)

Assume that $\lim_{x \to a} f(x) = L$ *and* $\lim_{x \to a} f(x) = M$. *Then* $L = M$.

Section 3

Arithmetic Rules for Limits of Functions

Connection with limits of sequences of numbers

Remark

The limit of $f(x)$ as x tends to c can be completely described in terms of the limits of sequences of numbers.

In fact, in evaluating $\lim_{x\to c} f(x)$, we often take a sequence of numbers $x_1, x_2, \ldots, x_n, \ldots$ that tend to c and we see whether the sequence $f(x_1), f(x_2), \ldots, f(x_n), \ldots$ tends to some number L.

In order for $\lim_{x\to c} f(x)$ to exist, we need to know that all sequences $\{x_i\}$ that tend to c result in sequences $\{f(x_i)\}$ that tend to L.

In one assignment problem, we ask you to explore the connection between the limit of a function at a point and limits of sequences of numbers. It will help you see why the next two theorems follow from the laws of arithmetic and the squeeze theorem for convergent sequences.

Theorem 2.3.1

If $\lim_{x\to c} f(x) = L_1, \lim_{x\to c} g(x) = L_2$, *and* $\lim_{x\to c} h(x) = L_3 \neq 0$, *then*

1. $\lim_{x\to c}(f(x) + g(x)) = L_1 + L_2$,
2. $\lim_{x\to c}(f(x)g(x)) = L_1 L_2$, *and*
3. $\lim_{x\to c} \frac{f(x)}{h(x)} = \frac{L_1}{L_3}$.

Limits of Polynomial Functions

Theorem 2.3.2 (Limits of Polynomial Functions)

Let $p(x)$ be any polynomial. That is,

$$p(x) = c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n$$

Then

$$\lim_{x \to a} p(x) = p(a)$$

Rational Function

A rational function is a function of the form

$$f(x) = \frac{P(x)}{Q(x)}$$

where $P(x)$ and $Q(x)$ are polynomials.

Problem: How do we find

$$\lim_{x \to a} f(x) = \lim_{x \to a} \frac{P(x)}{Q(x)}?$$

Example 2.3.3

Find $\lim_{x\to 1} \frac{x^2-2x}{x^2-x}$.

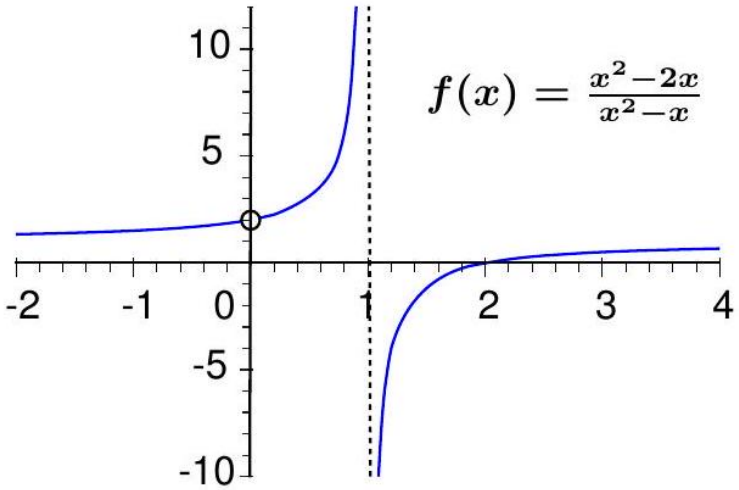
Solution: Note that

$$\lim_{x\to 1} x^2 - x = 0$$

but

$$\lim_{x\to 1} x^2 - 2x = -1 \neq 0$$

which
implies that $\lim_{x\to 1} \frac{x^2-2x}{x^2-x}$ does not exist.



Observation:
The graph of $f(x)$ shows that
the function is "unbounded" near $x = 1$.

Limits of Rational Functions

The following algorithm can be used to compute

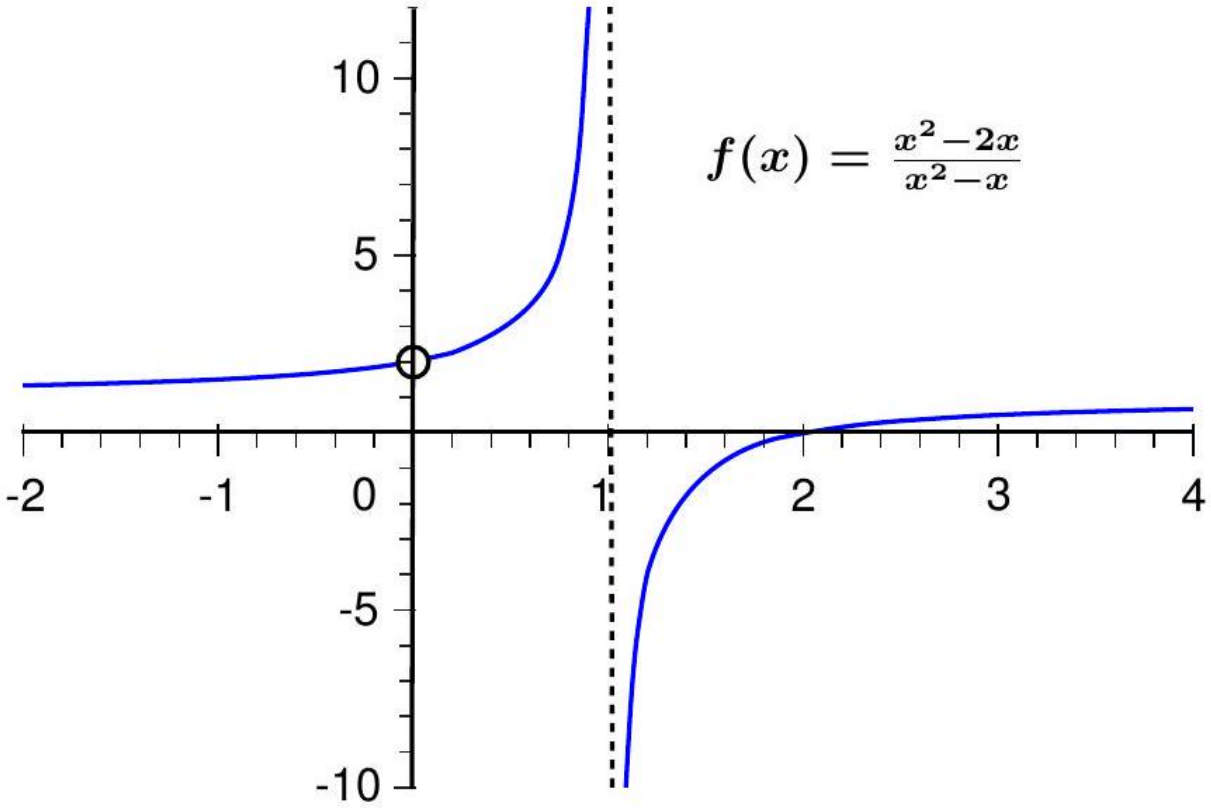
$$\lim_{x\to a} f(x) = \lim_{x\to a} \frac{P(x)}{Q(x)}$$

- Step 1) If $Q(a) \neq 0$, then $\lim_{x\to a} f(x) = \frac{\lim_{x\to a} P(x)}{\lim_{x\to a} Q(x)} = \frac{P(a)}{Q(a)} = f(a)$.
- Step 2) If $Q(a) = 0$ and $P(a) \neq 0$, then the limit does not exist.
- Step 3) If $Q(a) = 0 = P(a)$, then both $P(x)$ and $Q(x)$ can be factored into $Q(x) = (x-a)Q_1(x)$ and $P(x) = (x-a)P_1(x)$
  $\Rightarrow f(x) = \frac{(x-a)P_1(x)}{(x-a)Q_1(x)} = \frac{P_1(x)}{Q_1(x)} = f_1(x)$ for any $x \neq a$.

  Repeat Step 1) with $f_1(x) = \frac{P_1(x)}{Q_1(x)}$.

Examples

Recall: The graph of $f(x) = \frac{x^2-2x}{x^2-x} = \frac{P(x)}{Q(x)}$ appears as follows:



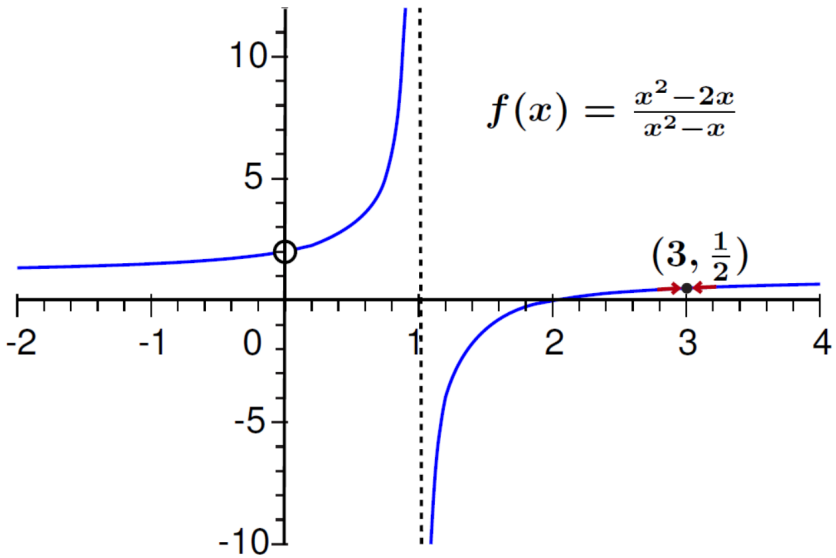
Examples

Example 2.3.4

Find $\lim_{x\to 3} f(x) = \lim_{x\to 3} \frac{x^2-2x}{x^2-x}$.

Solution: Note that

$$\begin{aligned} Q(3) &= (3)^2 - 3 \\ &= 6 \neq 0 \end{aligned}$$



Hence

$$\begin{aligned} \lim_{x\to 3} f(x) &= f(3) \\ &= \frac{3^2 - 2(3)}{3^2 - 3} \\ &= \frac{1}{2} \end{aligned}$$

Example

Example 2.3.5

Find $\lim_{x\to 1} \frac{x^2-2x}{x^2-x}$.

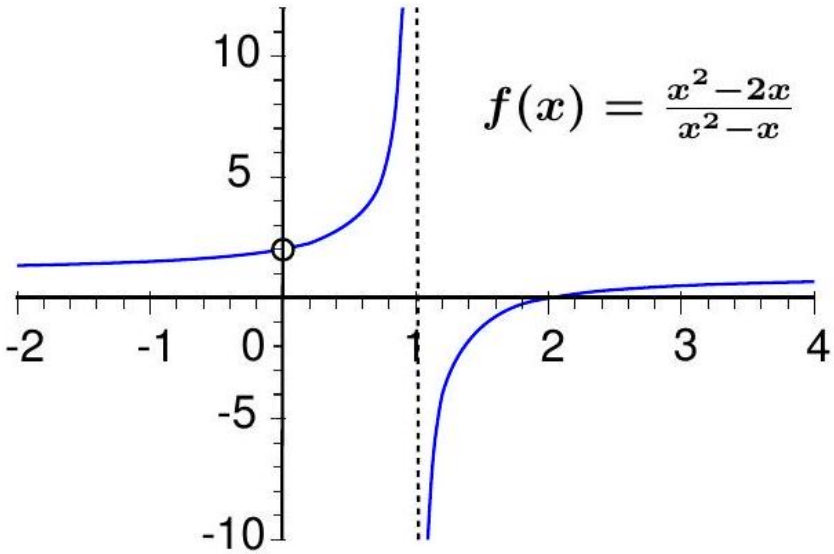
Solution: Since

$$Q(1) = 1^2 - 1 = 0$$

but

$$P(1) = 1^2 - 2(1) = -1 \neq 0$$

it follows that $\lim_{x\to 1} \frac{x^2-2x}{x^2-x}$ does not exist.



Examples

Example 2.3.6

Find $\lim_{x\to 0} \frac{x^2-2x}{x^2-x}$.

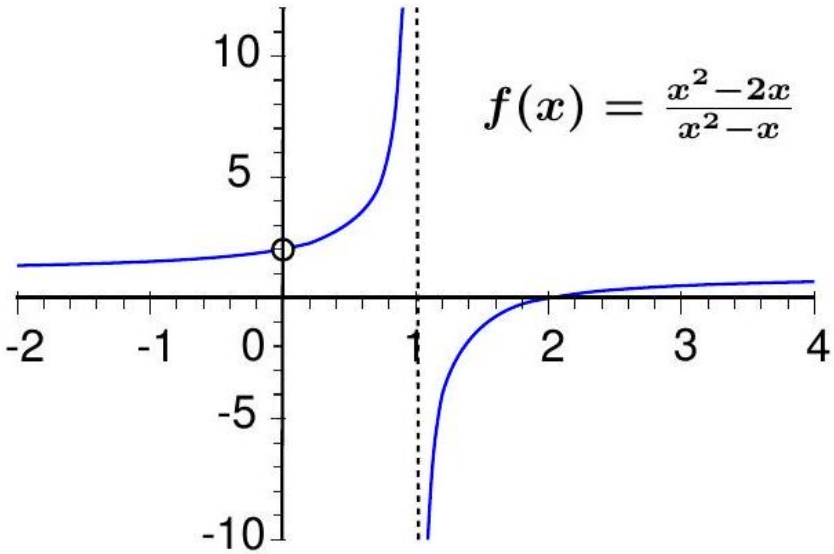
Solution: Note that

$$P(0) = 0 = Q(0)$$

So
both $P(x) = x^2 - x$ and $Q(x) = x^2 - 2x$
contain the factor $(x - 0) = x$.

Hence

$$\begin{aligned}\lim_{x\to 0} f(x) &= \lim_{x\to 0} \frac{(x)(x-2)}{(x)(x-1)} \\ &= \lim_{x\to 0} \frac{x-2}{x-1} \\ &= \frac{\lim_{x\to 0}(x-2)}{\lim_{x\to 0}(x-1)} \\ &= \frac{-2}{-1} = 2\end{aligned}$$



Squeeze Theorem for Limits

Theorem 2.3.7 (Squeeze Theorem for Limits)

Assume that $f(x), g(x)$, *and* $h(x)$ *are such that*

$$g(x) \leq f(x) \leq h(x)$$

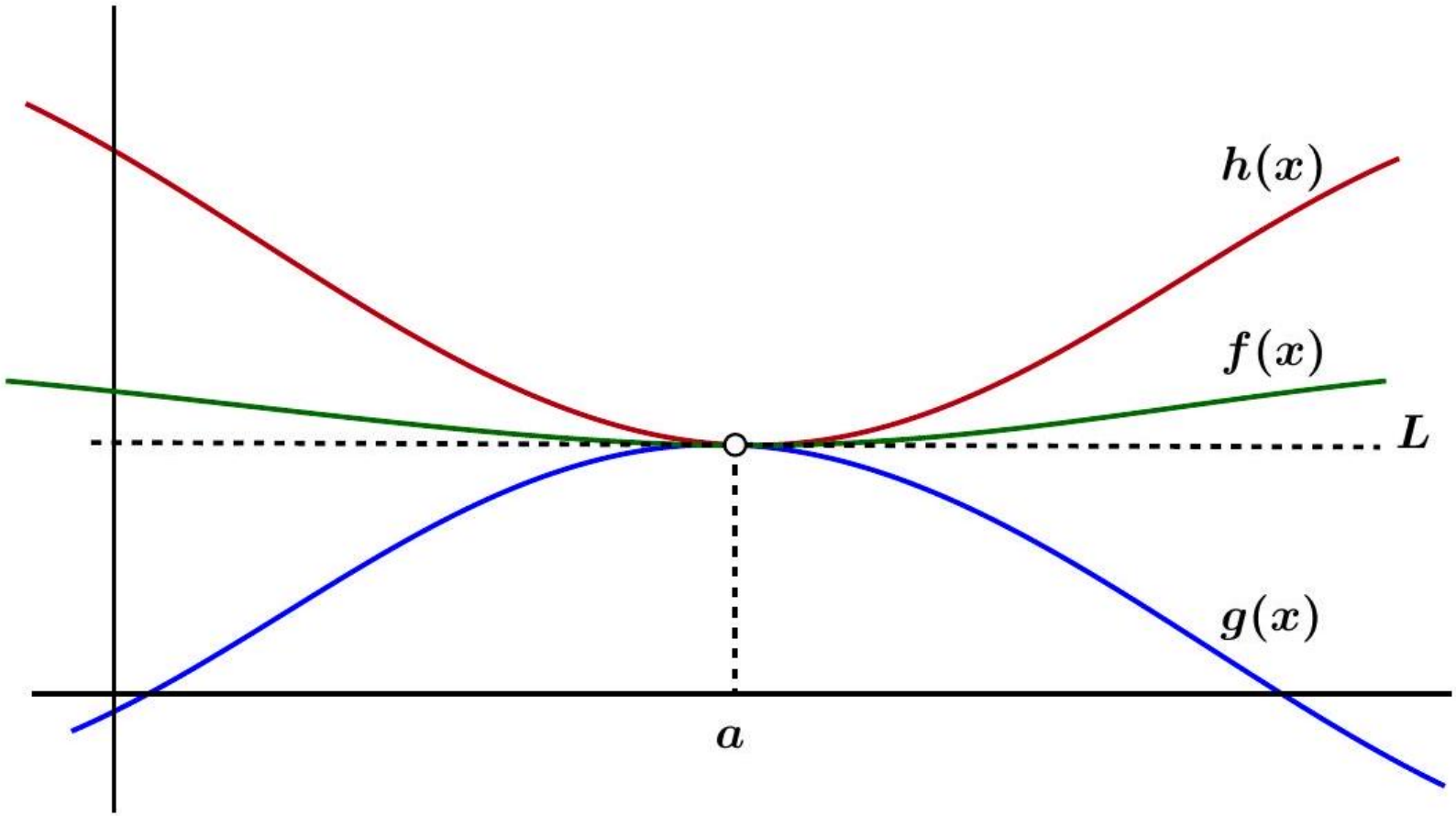
for all $x \neq a$.
Assume also that

$$\lim_{x \to a} g(x) = L = \lim_{x \to a} h(x)$$

Then $\lim_{x \to a} f(x)$ *exists and*

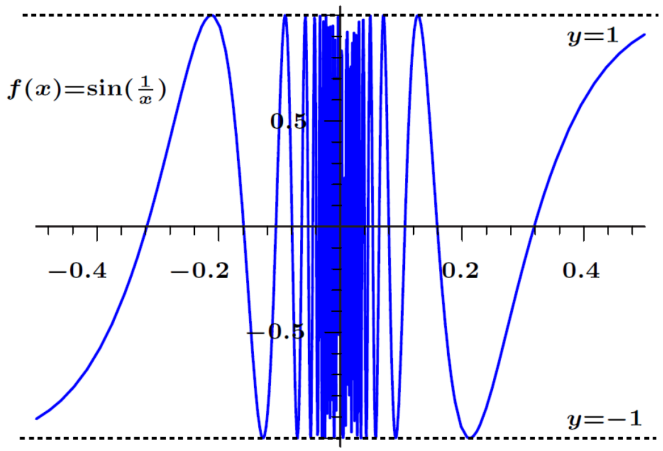
$$\lim_{x \to a} f(x) = L$$

Squeeze Theorem for Limits



Example

Let $f(x) = \sin(1/x)$.
We know that $\lim_{x \to 0} f(x)$ does not exist.

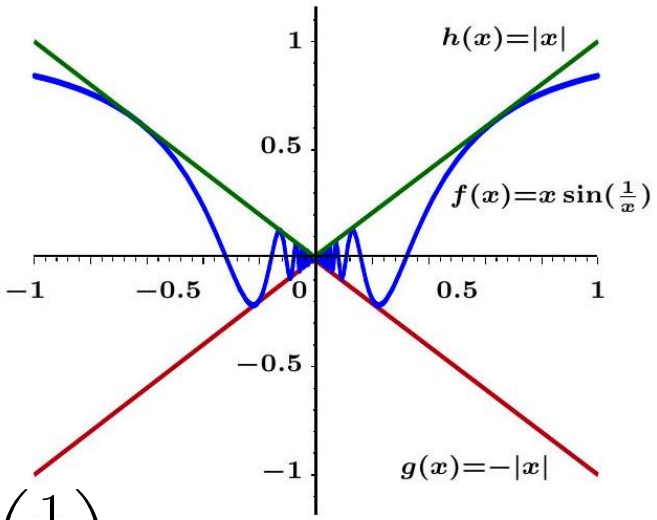


Example

Let $f(x) = \sin(1/x)$.

We know that $\lim_{x\to 0} f(x)$ does not exist.

Question:
What can we say about $\lim_{x\to 0} x \sin\left(\frac{1}{x}\right)$?



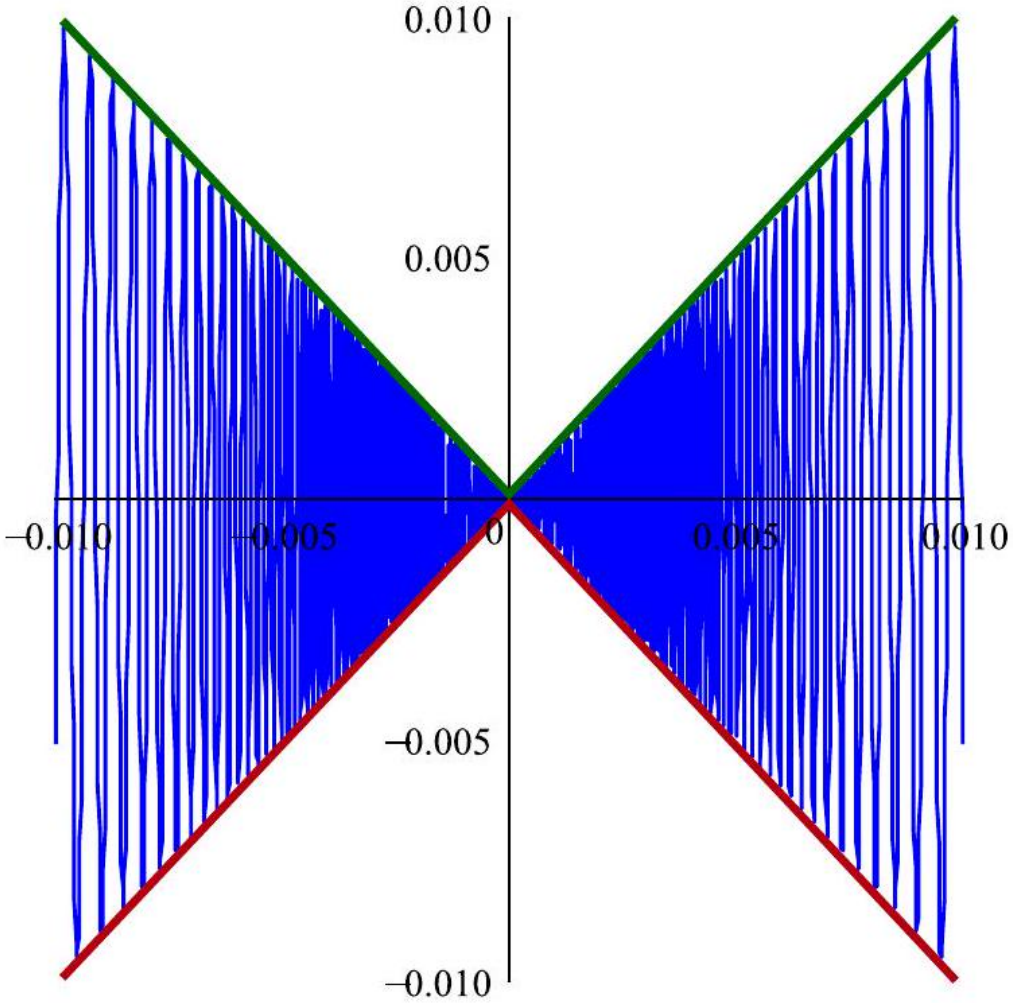
We know

$$\left|\sin\left(\frac{1}{x}\right)\right| \leq 1 \Rightarrow \left|x \sin\left(\frac{1}{x}\right)\right| \leq |x| \Rightarrow -|x| \leq x \sin\left(\frac{1}{x}\right) \leq |x|$$

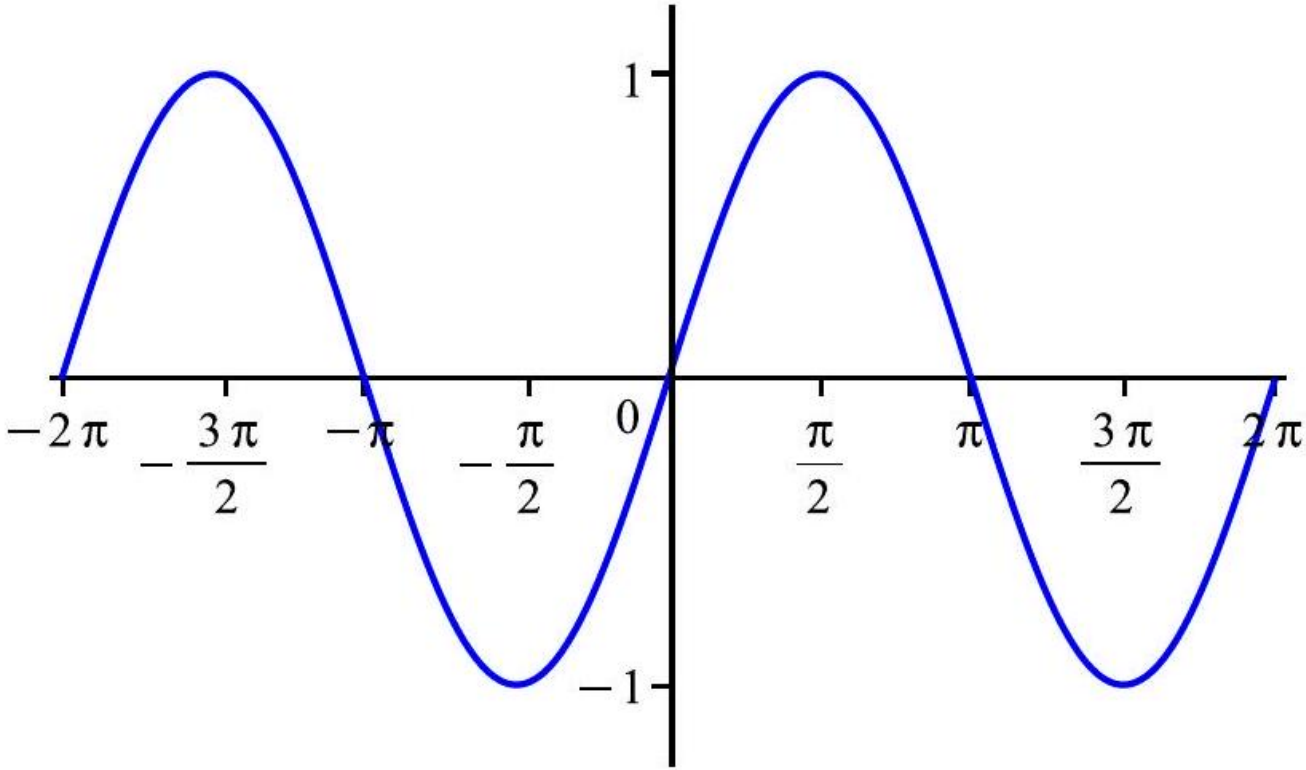
for all $\boldsymbol{x \neq 0}$. Hence,

$$\lim_{x\to 0} -|x| = 0 = \lim_{x\to 0} |x| \Rightarrow \lim_{x\to 0} x \sin\left(\frac{1}{x}\right) = 0$$

Example



$\lim_{\theta \to 0} \sin(\theta) = 0$



Remark

The graph of $\sin(\theta)$ suggests that

$$\lim_{\theta \to 0} \sin(\theta) = 0$$

The Squeeze Theorem can be used to confirm this limit.

$\lim_{\theta \to 0} \sin(\theta) = 0$

Observation: For

$$0 < \theta \leq \frac{\pi}{2}$$

we have

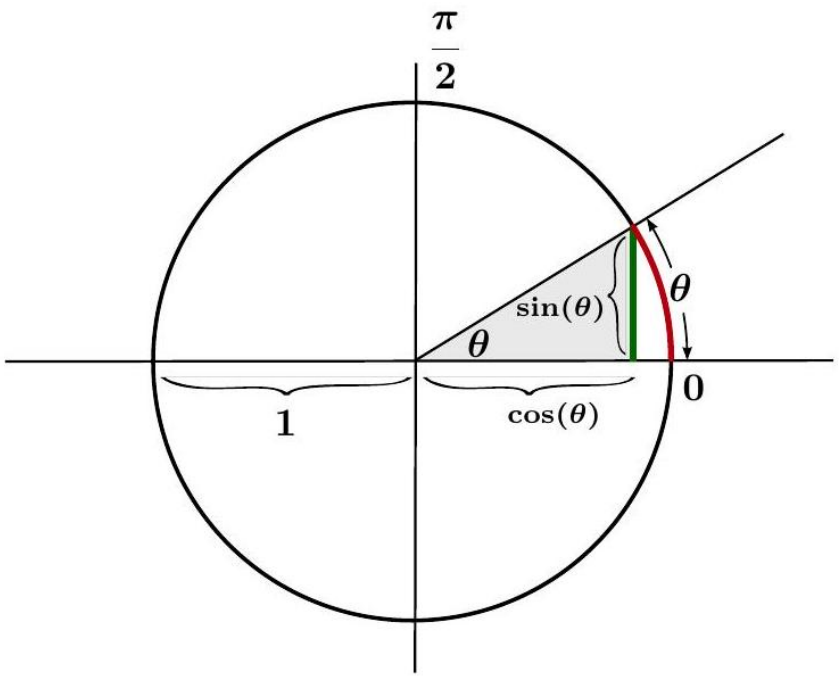
$$0 < \sin(\theta) < \theta$$

Since

$$\lim_{\theta \to 0^+} 0 = 0 = \lim_{\theta \to 0^+} \theta$$

then

$$\lim_{\theta \to 0^+} \sin(\theta) = 0$$



$\lim_{\theta \to 0} \sin(\theta) = 0$

Observation (continued): Let $\frac{-\pi}{2} < \theta < 0$. Since $\sin(\theta)$ is an odd function,

$$\sin(-\theta) = -\sin(\theta)$$

and since if $\theta \to 0^-$, we have $0 < -\theta < \frac{\pi}{2}$ and $-\theta \to 0^+$,

$$\begin{aligned}
\lim_{\theta \to 0^-} \sin(\theta) &= \lim_{\theta \to 0^-} -\sin(-\theta) \\
&= \lim_{(-\theta) \to 0^+} -\sin(-\theta) \\
&= -\lim_{(-\theta) \to 0^+} \sin(-\theta) \\
&= -1 \cdot 0 \\
&= 0.
\end{aligned}$$

Finally,

$$\lim_{\theta \to 0^-} \sin(\theta) = 0 = \lim_{\theta \to 0^+} \sin(\theta)$$

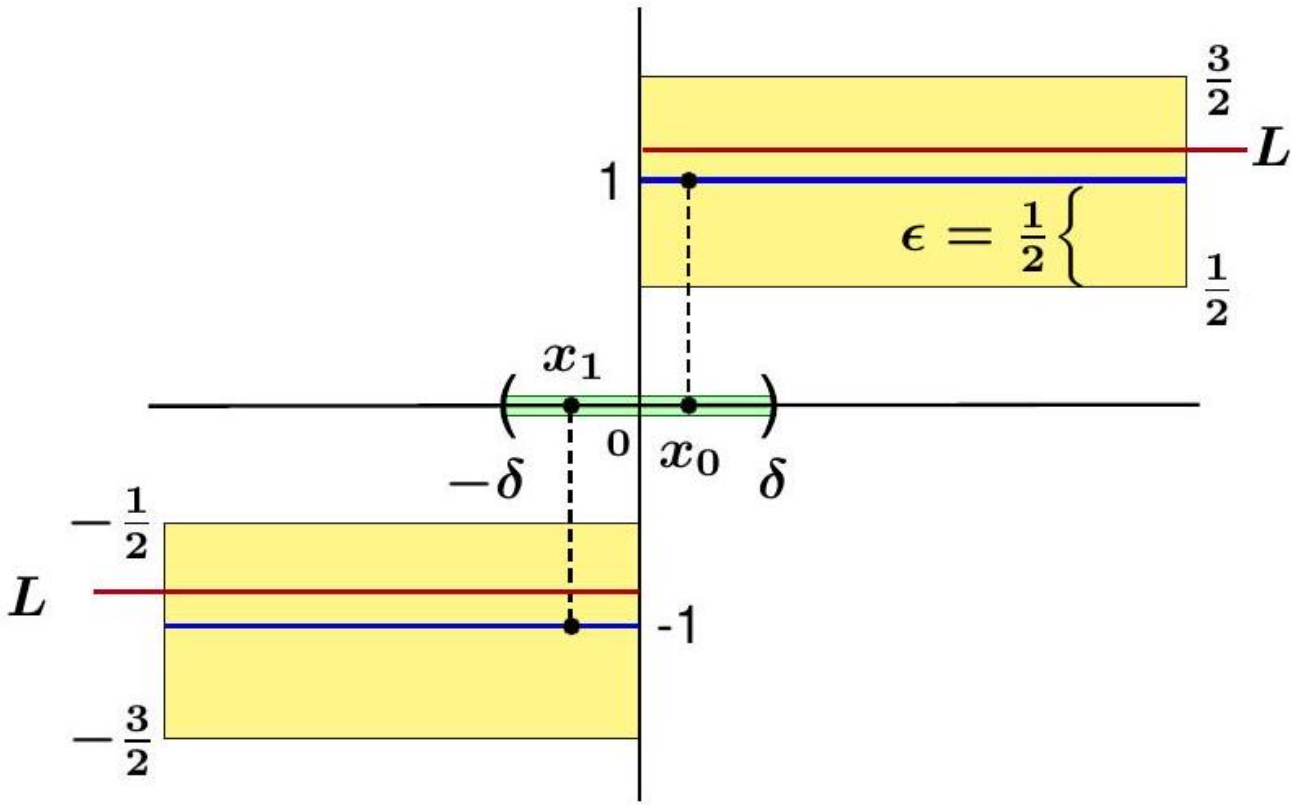
So

$$\lim_{\theta \to 0} \sin(\theta) = 0$$

Section 4

One-sided Limits and Limits at Infinity

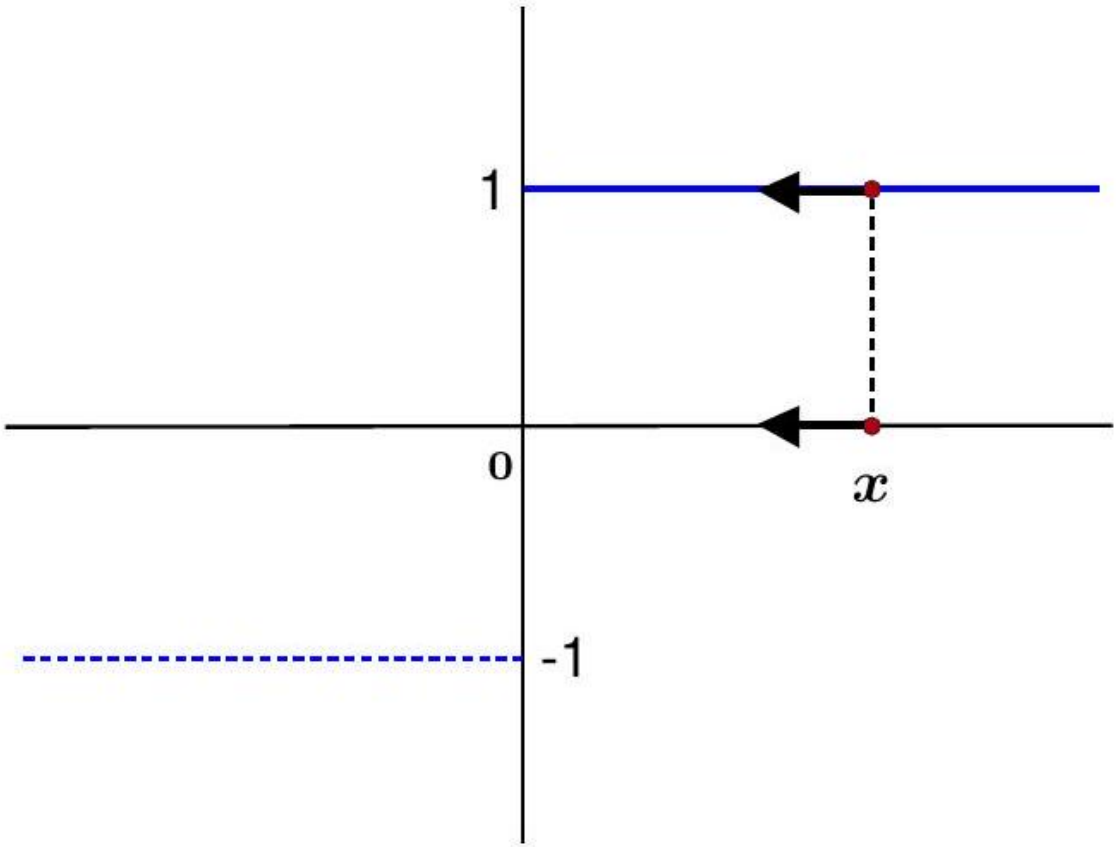
Example



Example

Let $f(x) = \frac{|x|}{x}$. $\Rightarrow \lim_{x \to 0} f(x)$ does not exist.

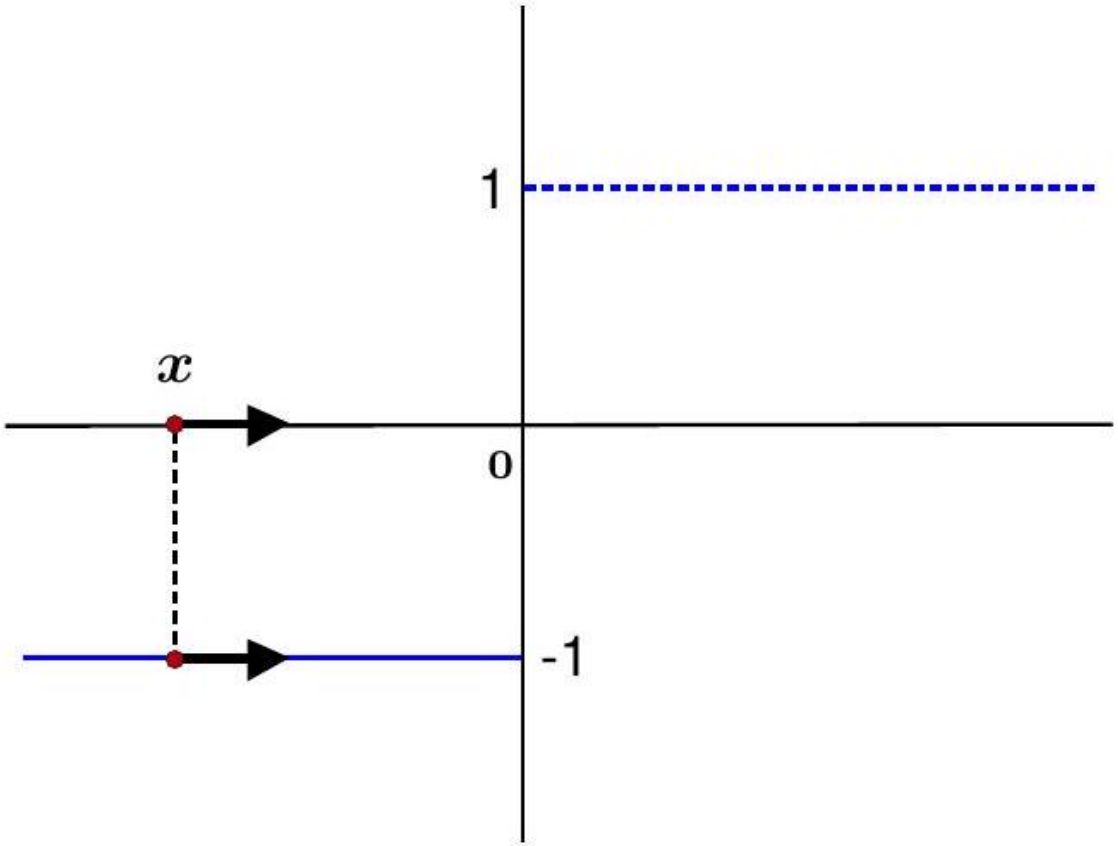
Example



Example

Let $f(x) = \frac{|x|}{x}$. $\Rightarrow \lim_{x \to 0} f(x)$ does not exist.
If $x > 0$ and $x \to 0$, then $f(x) \to 1$.

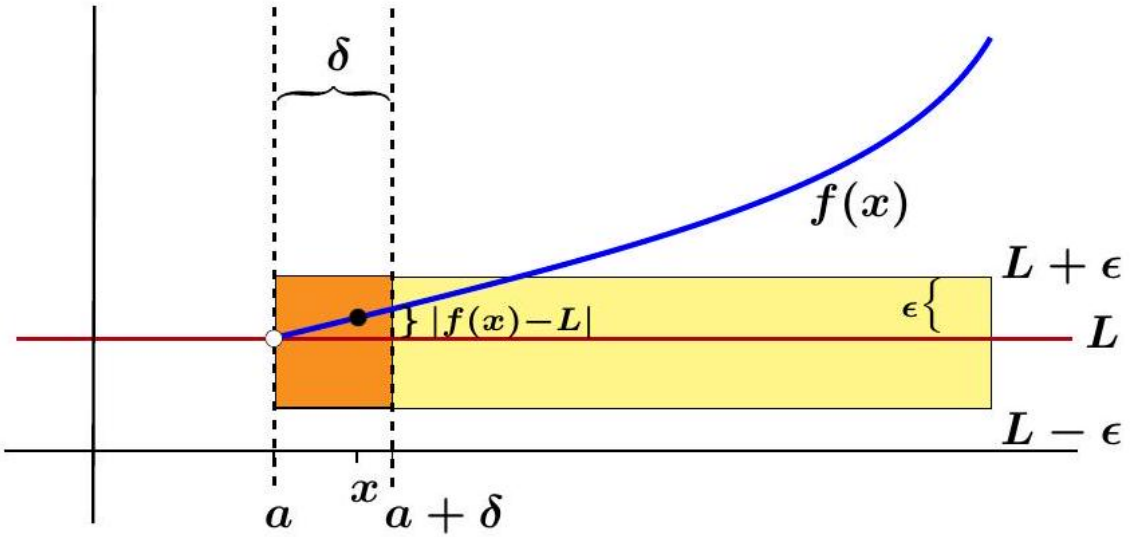
Example



Example

Let $f(x) = \frac{|x|}{x}$. $\Rightarrow \lim_{x \to 0} f(x)$ does not exist.
If $x > 0$ and $x \to 0$, then $f(x) \to 1$.
If $x < 0$ and $x \to 0$, then $f(x) \to -1$.

One-sided Limits



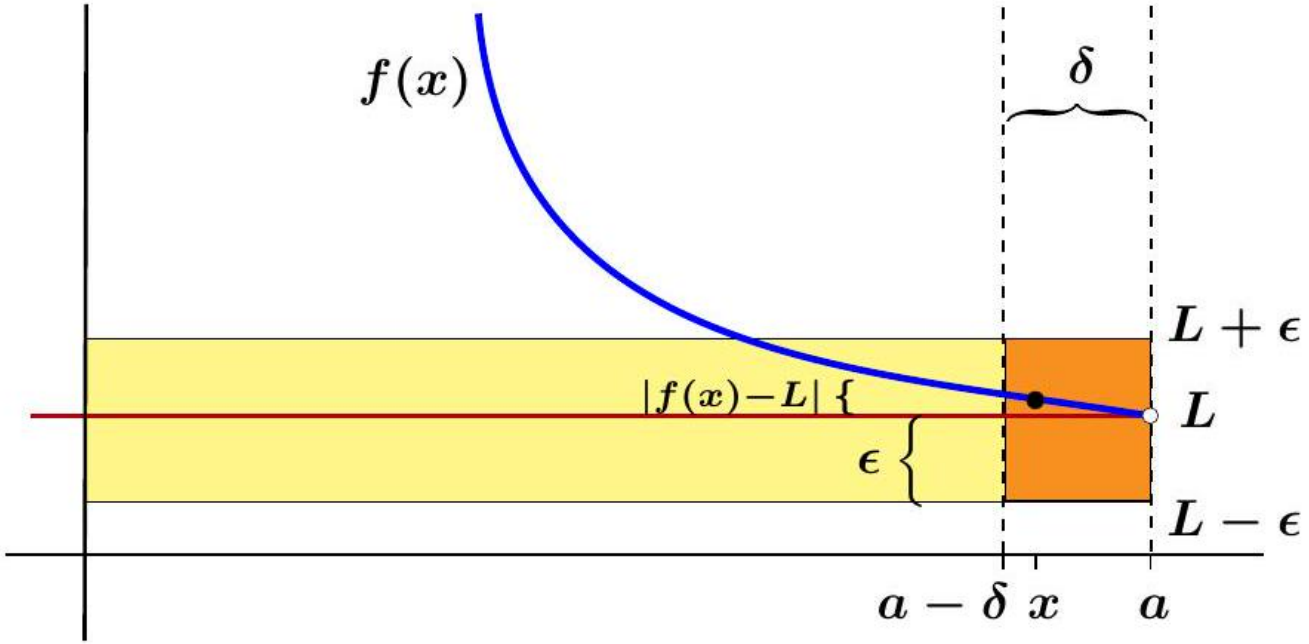
Definition 2.4.1 (Limit from Above)

We say that L is the limit of $f(x)$ as x approaches a from above (or from the right), if for every $\epsilon > 0$ there exists a $\delta > 0$ such that if $a < x < a + \delta$, then

$$|f(x) - L| < \epsilon$$

We write $\lim_{x \to a^+} f(x) = L$.

One-sided Limits



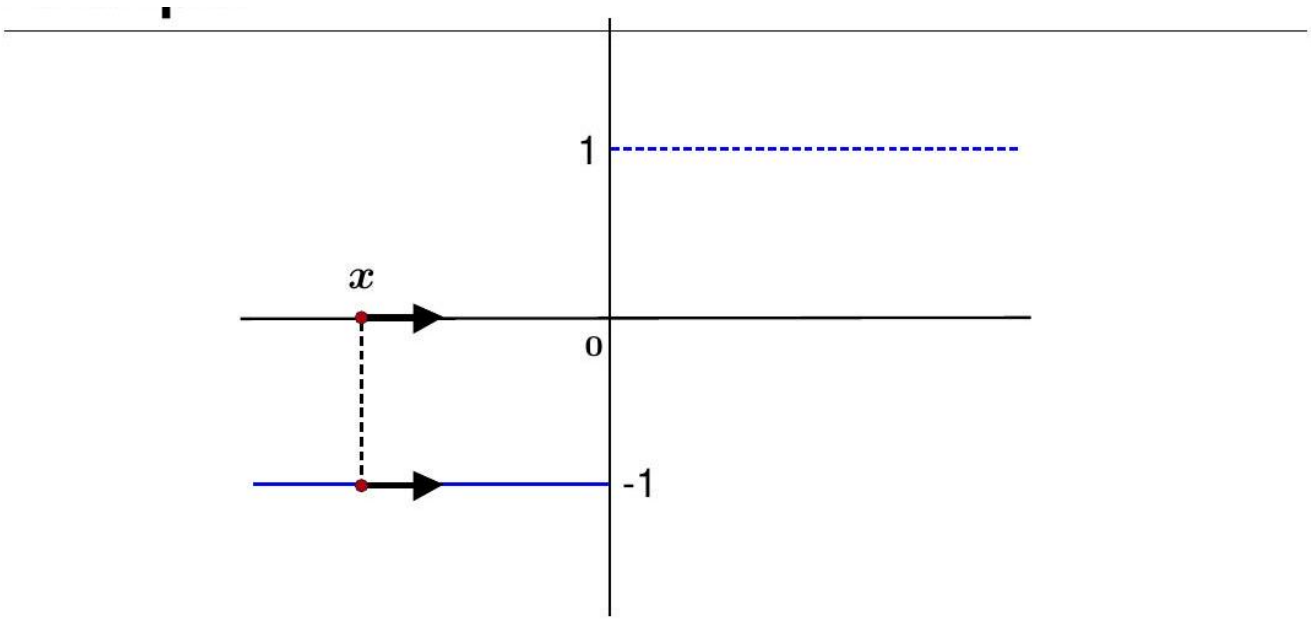
Definition 2.4.2 (Limit from Below)

We say that L is the limit of $f(x)$ as x approaches a from below (or from the left), if for every $\epsilon > 0$ there exists a $\delta > 0$ such that if $a - \delta < x < a$, then

$$|f(x) - L| < \epsilon$$

We write $\lim_{x \to a^-} f(x) = L$.

Example



Example 2.4.3

Example: We know that $\lim_{x\to 0} \frac{|x|}{x}$ does not exist. However,

$$\lim_{x\to 0^+} \frac{|x|}{x} = 1$$

and

$$\lim_{x\to 0^-} \frac{|x|}{x} = -1$$

One-sided Limits vs. Limits

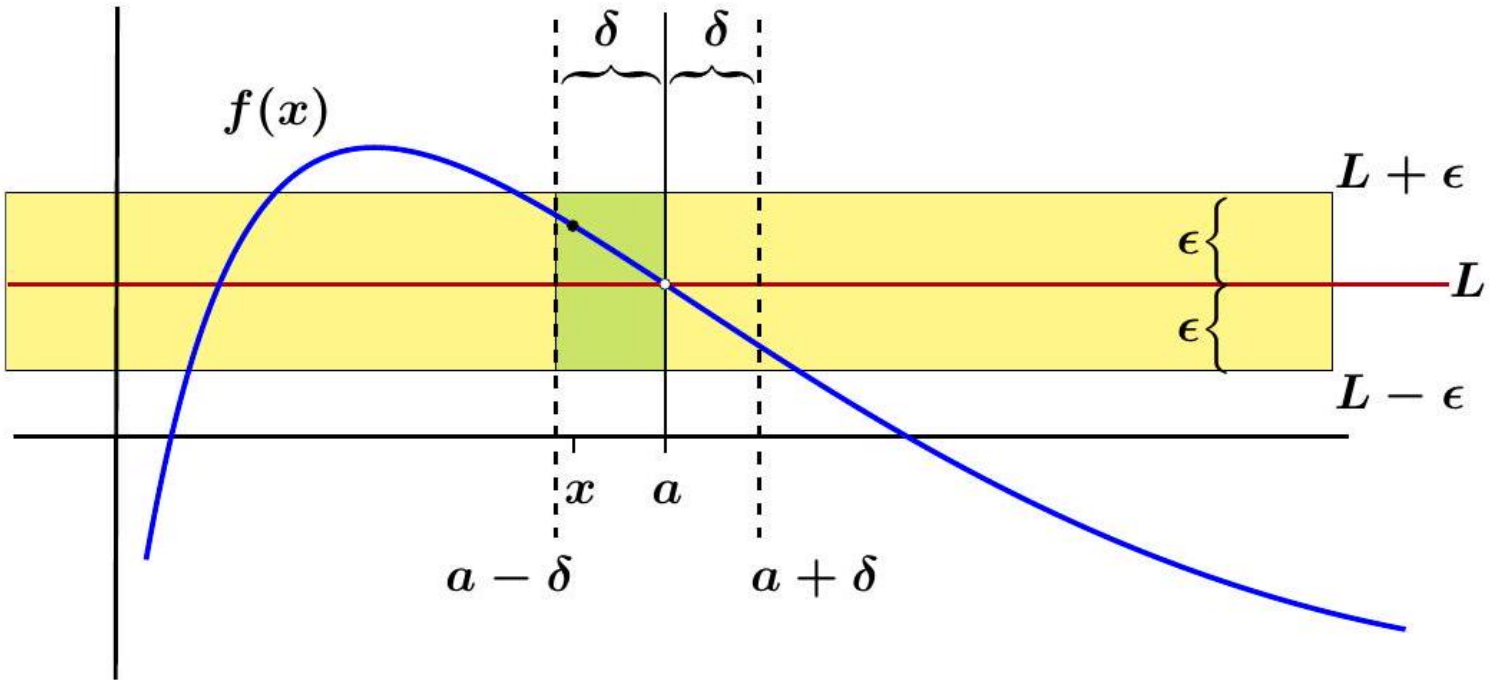
Theorem 2.4.4

The following are equivalent:

1. $\lim_{x \to a} f(x) = L$.
2. *Both* $\lim_{x \to a^+} f(x)$ *and* $\lim_{x \to a^-} f(x)$ *exist with*

$$\lim_{x \to a^+} f(x) = L = \lim_{x \to a^-} f(x)$$

One-sided Limits

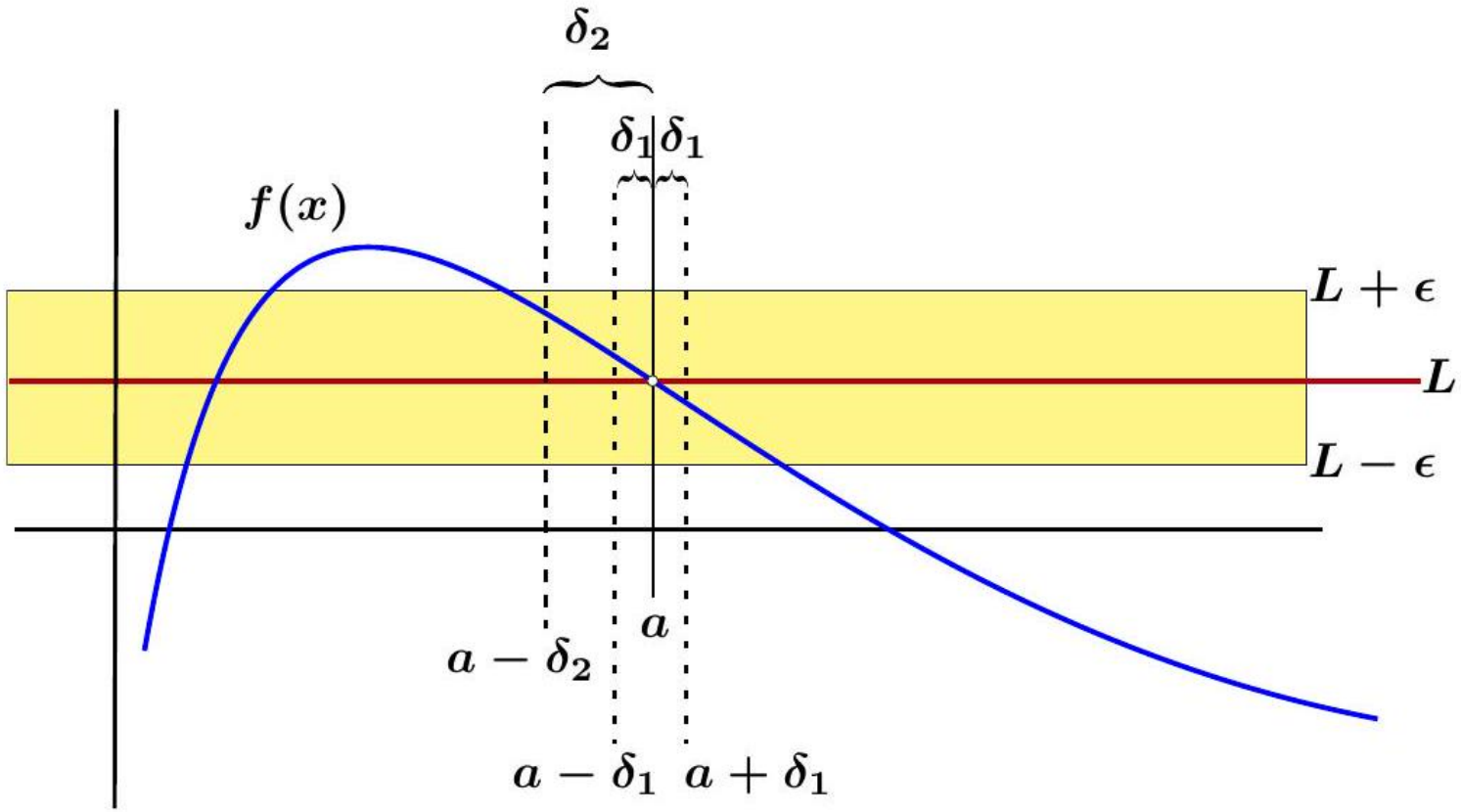


Assume that $\lim_{x\to a} f(x) = L$

$$\Rightarrow \quad \lim_{x\to a^+} f(x) = L, \text{ and}$$

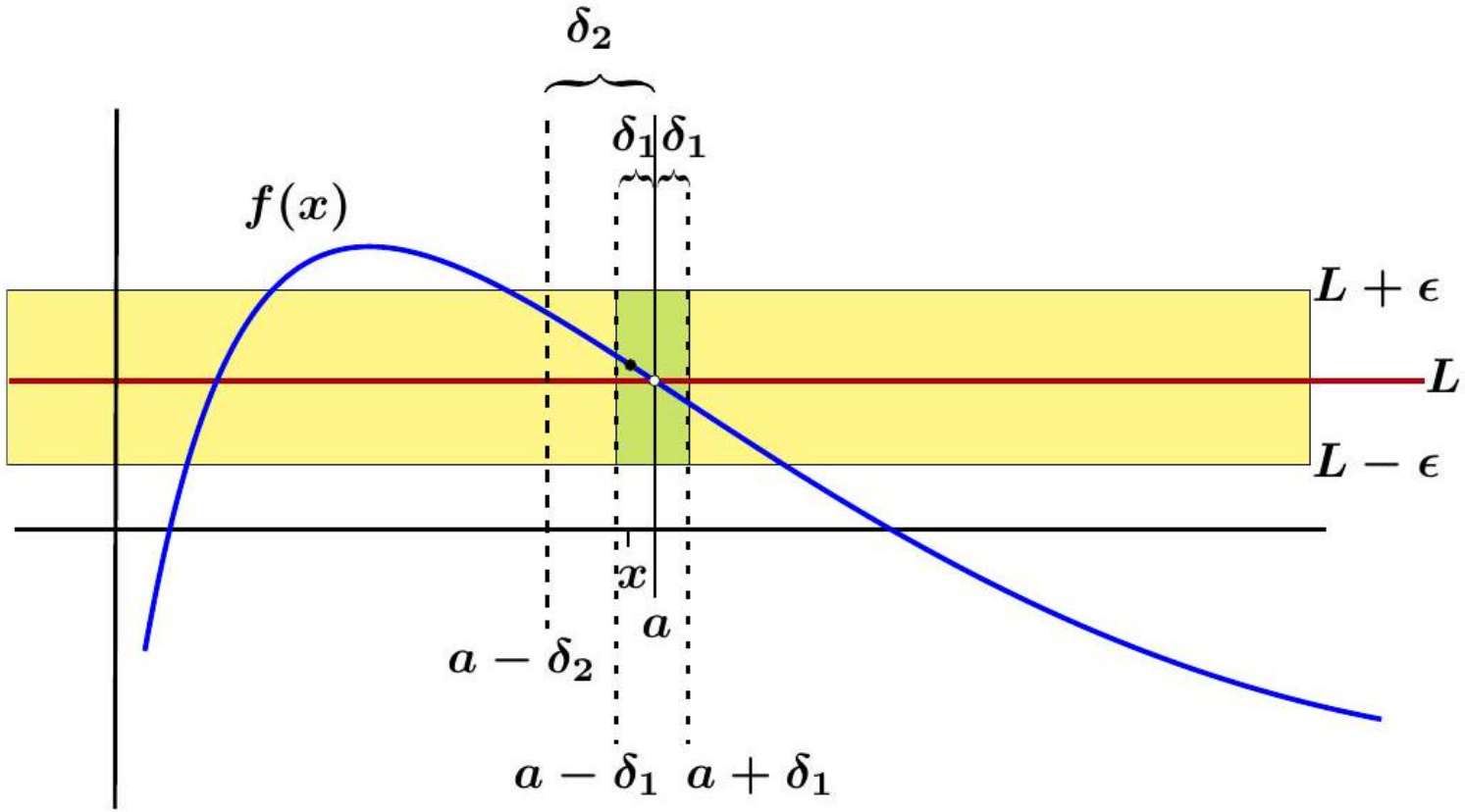
$$\lim_{x\to a^-} f(x) = L$$

One-sided Limits



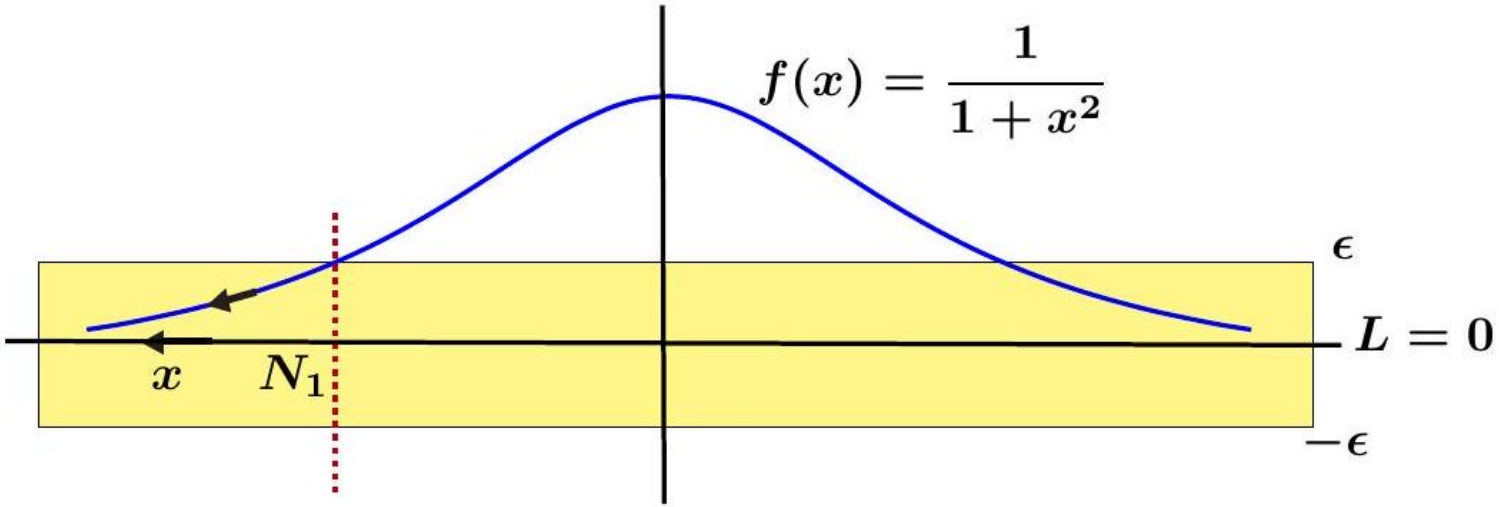
Assume that $\lim_{x\to a^+} f(x) = L = \lim_{x\to a^-} f(x)$

One-sided Limits



Assume that $\lim_{x\to a^+} f(x) = L = \lim_{x\to a^-} f(x) \Rightarrow \lim_{x\to a} f(x) = L$.

Limits at ∞

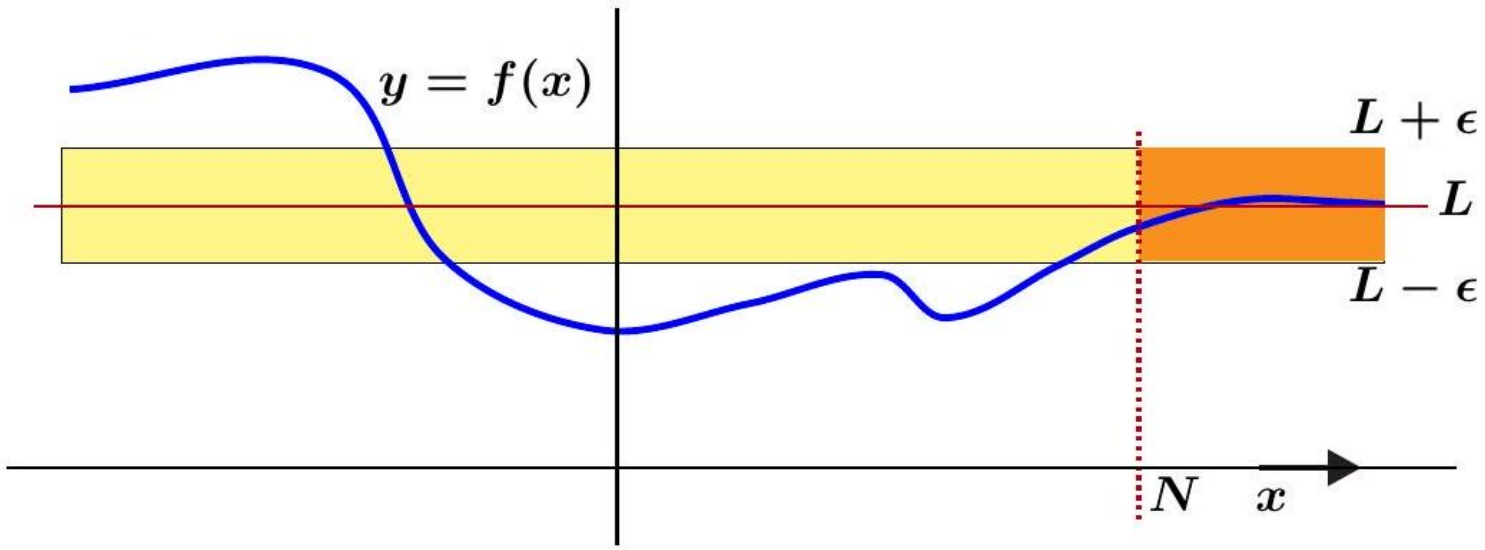


Observation: As x approaches $-\infty, 1 + x^2$ becomes very large and hence $f(x) = \frac{1}{1+x^2}$ gets very close to 0 .

More precisely, given any positive tolerance $\epsilon > 0$, we can find a cutoff N_1 so that if $x < N_1$, then $f(x)$ approximates 0 with an error less than ϵ.

We want to say that 0 is the limit as x approaches $-\infty$ of $f(x)$.

Limits at ∞



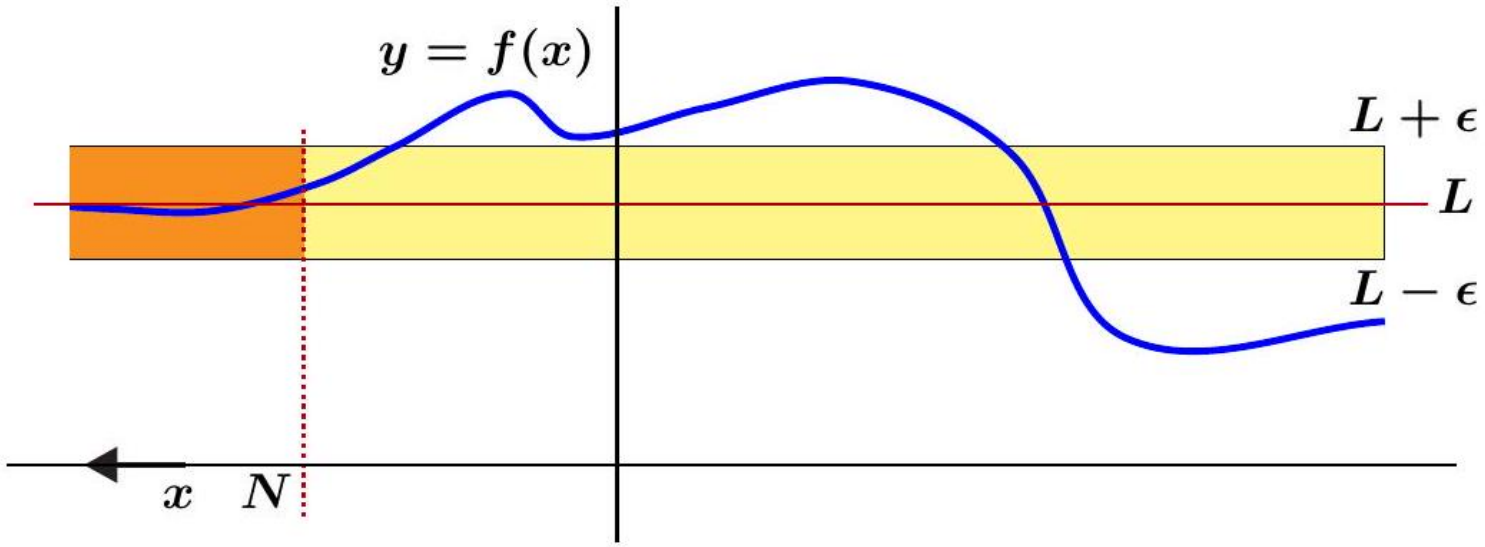
Definition 2.4.5 (Limit at ∞)

We say that L is the limit of $f(x)$ as x approaches ∞ if for every $\epsilon > 0$ there exists a cutoff $N > 0$ such that if $x > N$, then

$$|f(x) - L| < \epsilon$$

We write $\lim_{x \to \infty} f(x) = L$.

Limits at ∞



Definition 2.4.6 (Limit at $-\infty$)

We say that L is the limit of $f(x)$ as x approaches $-\infty$ if for every $\epsilon > 0$ there exists a cutoff $N < 0$ such that if $x < N$, then

$$|f(x) - L| < \epsilon$$

We write $\lim_{x \to -\infty} f(x) = L$.

Rational Functions

Example 2.4.7

Evaluate

$$\lim_{x\to\infty} \frac{2x^2-3x+4}{x^2+x-5}$$

Observation: For polynomials and large values of x, the highest power terms dominate. Hence, for x is very large

$$\frac{2x^2-3x+4}{x^2+x-5} \cong \frac{2x^2}{x^2} = 2$$

We might guess that

$$\lim_{x\to\infty} \frac{2x^2-3x+4}{x^2+x-5} = 2$$

Example 2.4.8 (continued)

Evaluate

$$\lim_{x\to\infty} \frac{2x^2-3x+4}{x^2+x-5}$$

Solution: We have for all $x > 0$,

$$\begin{aligned}\frac{2x^2-3x+4}{x^2+x-5} &= \frac{x^2\left(2-\frac{3}{x}+\frac{4}{x^2}\right)}{x^2\left(1+\frac{1}{x}-\frac{5}{x^2}\right)}\\ &= \frac{2-\frac{3}{x}+\frac{4}{x^2}}{1+\frac{1}{x}-\frac{5}{x^2}}\end{aligned}$$

Then

$$\begin{aligned}\lim_{x\to\infty} \frac{2x^2-3x+4}{x^2+x-5} &= \lim_{x\to\infty} \frac{2-\frac{3}{x}+\frac{4}{x^2}}{1+\frac{1}{x}-\frac{5}{x^2}}\\ &= \frac{2-0+0}{1+0-0}\\ &= 2\end{aligned}$$

Rational Functions

Example 2.4.9

Evaluate

$$\lim_{x\to-\infty}\frac{3x^3+x}{x^4+1}$$

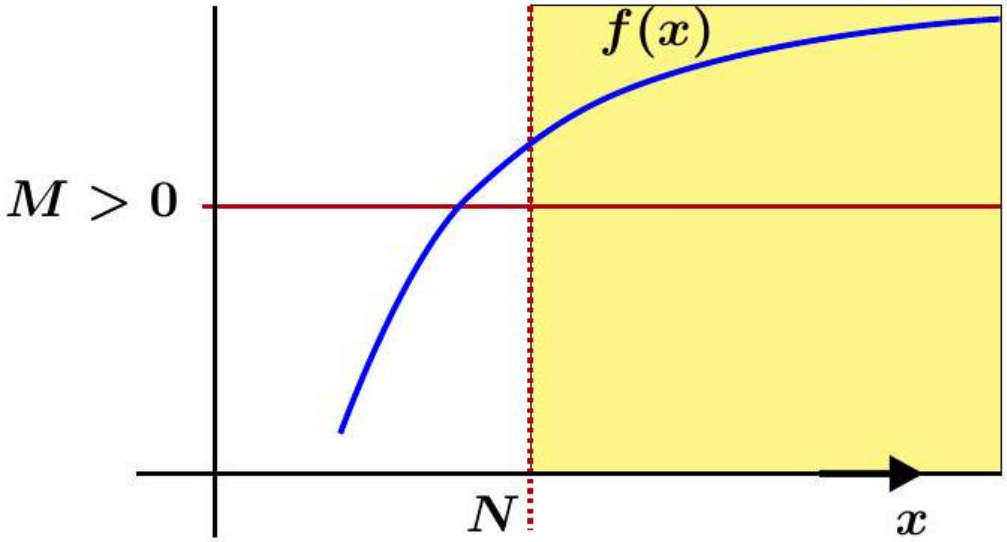
Solution: We have

$$\begin{aligned}\lim_{x\to-\infty}\frac{3x^3+x}{x^4+1} &= \lim_{x\to-\infty}\frac{x^3\left(3+\frac{1}{x^2}\right)}{x^4\left(1+\frac{1}{x^4}\right)}\\ &= \lim_{x\to-\infty}\left(\frac{1}{x}\right)\left(\frac{3+\frac{1}{x^2}}{1+\frac{1}{x^4}}\right)\\ &= 0\cdot 3\end{aligned}$$

Hence

$$\lim_{x\to-\infty}\frac{3x^3+x}{x^4+1}=0$$

Infinite Limits at ∞



Definition 2.4.10 (Infinite Limits at ∞)

We say that the limit of $f(x)$ as x approaches ∞ is ∞ if for every $M > 0$ there exists a cutoff $N > 0$ such that if $x > N$, then

$$f(x) > M$$

We write

$$\lim_{x \to \infty} f(x) = \infty$$

Note: We can define $\lim_{x \to \infty} f(x) = -\infty$ and $\lim_{x \to -\infty} f(x) = \pm\infty$ similarly.

Limits at ∞

Example 2.4.11

Evaluate

$$\lim_{x\to\infty} \frac{3x^3+x}{x^2+1}$$

Following the same procedure as before, we have

$$\begin{aligned}\frac{3x^3+x}{x^2+1} &= \frac{x^3\left(3+\frac{1}{x^2}\right)}{x^2\left(1+\frac{1}{x^2}\right)} \\ &= x\cdot\left(\frac{3+\frac{1}{x^2}}{1+\frac{1}{x^2}}\right) \\ &\cong 3x\end{aligned}$$

Hence

$$\lim_{x\to\infty} \frac{3x^3+x}{x^2+1} = \infty$$

Note: Formally, this means that the limit does not exist.

Limits of the Form $\lim_{x\to\pm\infty} \frac{p(x)}{q(x)}$

Let

$$f(x) = \frac{b_0 + b_1x + b_2x^2 + b_3x^3 + \cdots + b_jx^j}{c_0 + c_1x + c_2x^2 + \cdots + c_kx^k}$$

By factoring out x^j from the numerator and x^k from the denominator and rewriting the function as

$$f(x) = \frac{x^j}{x^k}\left[\frac{\frac{b_0}{x^j} + \frac{b_1}{x^{j-1}} + \frac{b_2}{x^{j-2}} + \frac{b_3}{x^{j-3}} + \cdots + b_j}{\frac{c_0}{x^k} + \frac{c_1}{x^{k-1}} + \frac{c_2}{x^{k-2}} + \cdots + c_k}\right]$$

This leads us to conclude that

$$\lim_{x\to\pm\infty} \frac{p(x)}{q(x)} = \begin{cases} \frac{b_j}{c_k} & \text{if } j = k \\ 0 & \text{if } j < k \\ \pm\infty & \text{if } j > k \end{cases}$$

where in the last case whether the limit is ∞ or $-\infty$ depends on the sign of $\frac{b_j}{c_k}$ and whether $x \to \infty$ or $-\infty$, as well as whether or not $j - k$ is even or odd.

Squeeze Theorem for Limits at $\pm\infty$

Theorem 2.4.12 (Squeeze Theorem for Limits at $\pm\infty$)

Assume that $g(x) \leq f(x) \leq h(x)$ *for all* $x \geq N$. *If*

$$\lim_{x\to\infty} g(x) = L = \lim_{x\to\infty} h(x)$$

then $\lim_{x\to\infty} f(x)$ *exists and it equals* L.
Assume that $g(x) \leq f(x) \leq h(x)$ *for all* $x \leq N$. *If*

$$\lim_{x\to-\infty} g(x) = L = \lim_{x\to-\infty} h(x)$$

then $\lim_{x\to-\infty} f(x)$ *exists and it equals* L.

Limits of the form $\lim_{x\to\pm\infty} \frac{\sin(x)}{x}$

Example 2.4.13

Evaluate

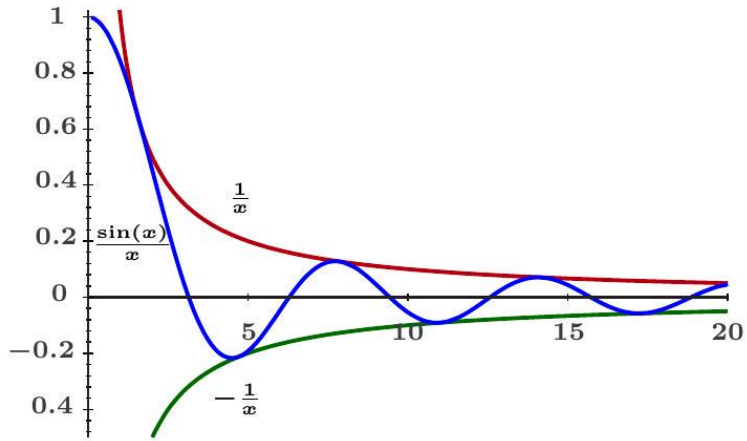
$$\lim_{x\to\infty} \frac{\sin(x)}{x}$$

Solution: Since

$$|\sin(x)| \leq 1$$

for any $x > 0$, we have

$$-\frac{1}{x} \leq \frac{\sin(x)}{x} \leq \frac{1}{x}$$



Now

$$\lim_{x\to\infty} -\frac{1}{x} = 0 = \lim_{x\to\infty} \frac{1}{x}$$

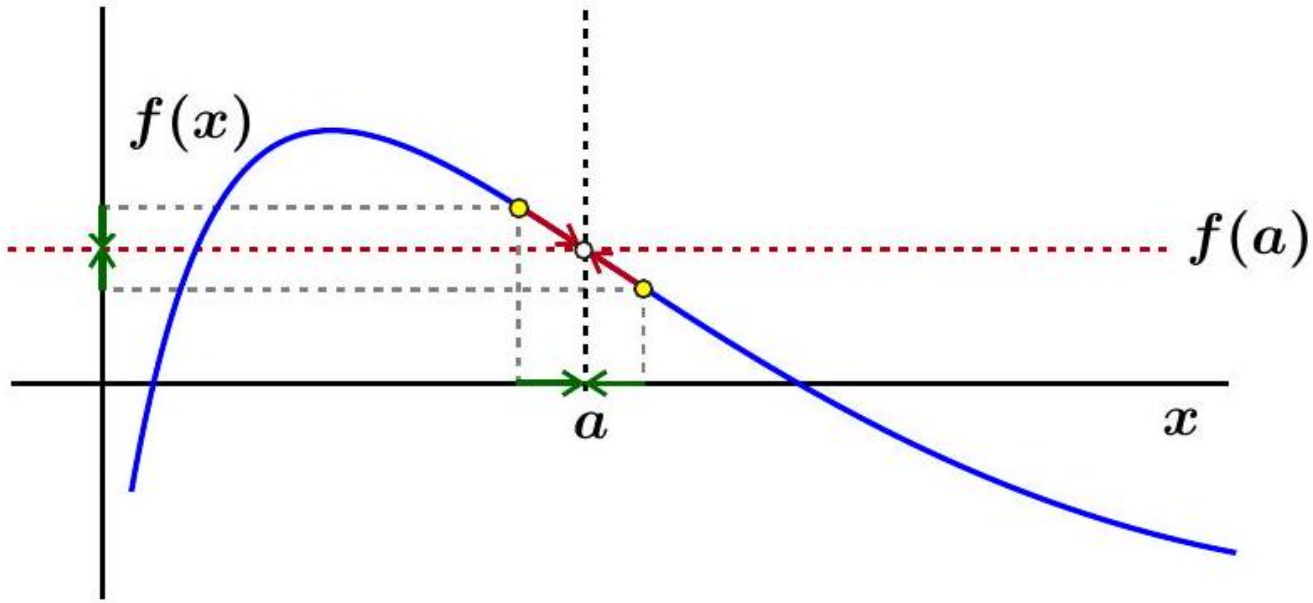
so the Squeeze Theorem implies that

$$\lim_{x\to\infty} \frac{\sin(x)}{x} = 0$$

Section 5

Continuity

Definition of Continuity



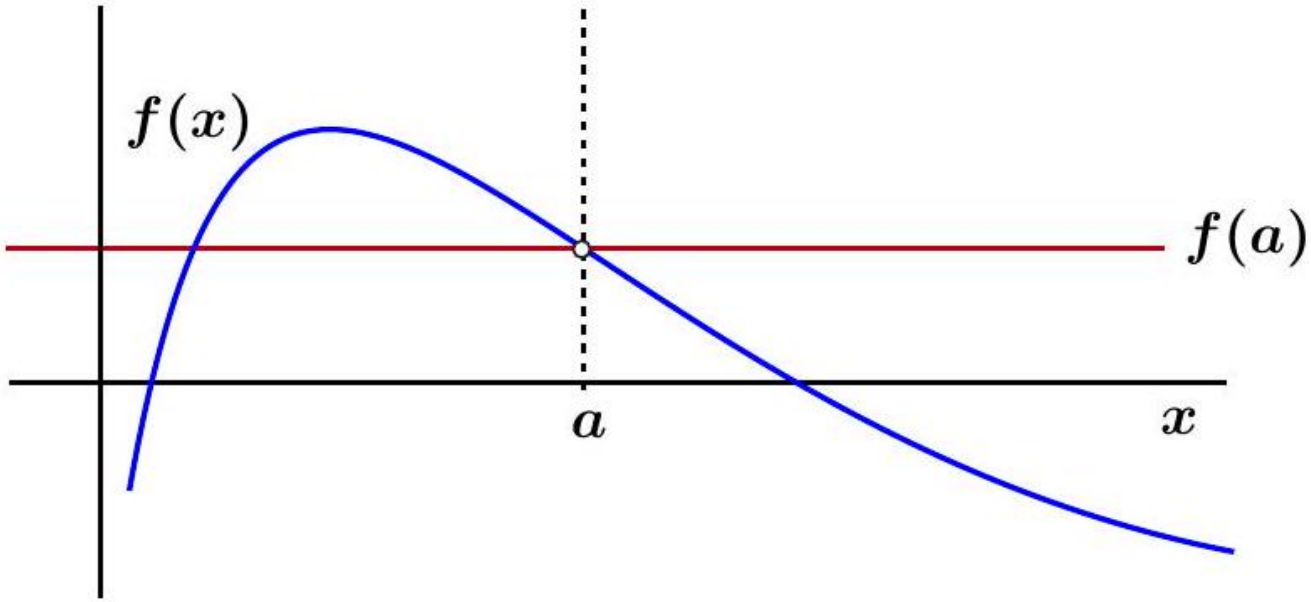
Definition 2.5.1 (Continuity)

We say that $f(x)$ is continuous at $x = a$ if

1. $\lim_{x \to a} f(x)$ exists.
2. $\lim_{x \to a} f(x) = f(a)$.

Otherwise, we say that $f(x)$ is discontinuous at a or that a is a point of discontinuity for $f(x)$.

Definition of Continuity



Definition 2.5.1 (Continuity)

We say that $f(x)$ is continuous at $x = a$ if

1. $\lim_{x \to a} f(x)$ exists.
2. $\lim_{x \to a} f(x) = f(a)$.

Otherwise, we say that $f(x)$ is discontinuous at a or that a is a point of discontinuity for $f(x)$.

Note: $2) \Rightarrow 1$).

Definition of Continuity

Observe: The following are equivalent:

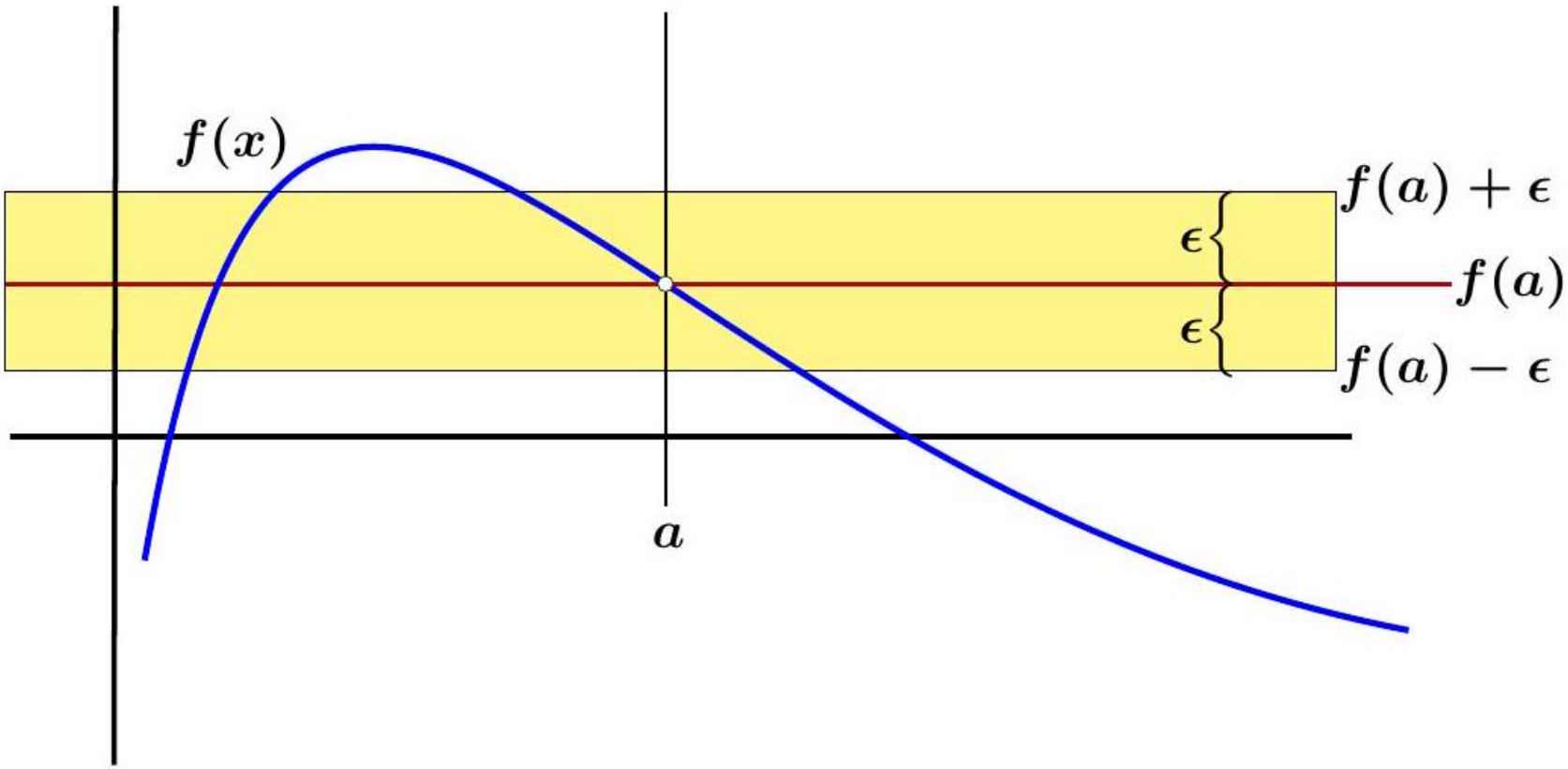
1. $\lim_{x \to a} f(x) = f(a)$.
2. For every $\epsilon > 0$, there exists a $\delta > 0$ such that $0 < |x - a| < \delta \Rightarrow |f(x) - f(a)| < \epsilon$.
3. For every $\epsilon > 0$, there exists a $\delta > 0$ such that $|x - a| < \delta \Rightarrow |f(x) - f(a)| < \epsilon$.

Definition 2.5.2 (Alternate Definition: Continuity)

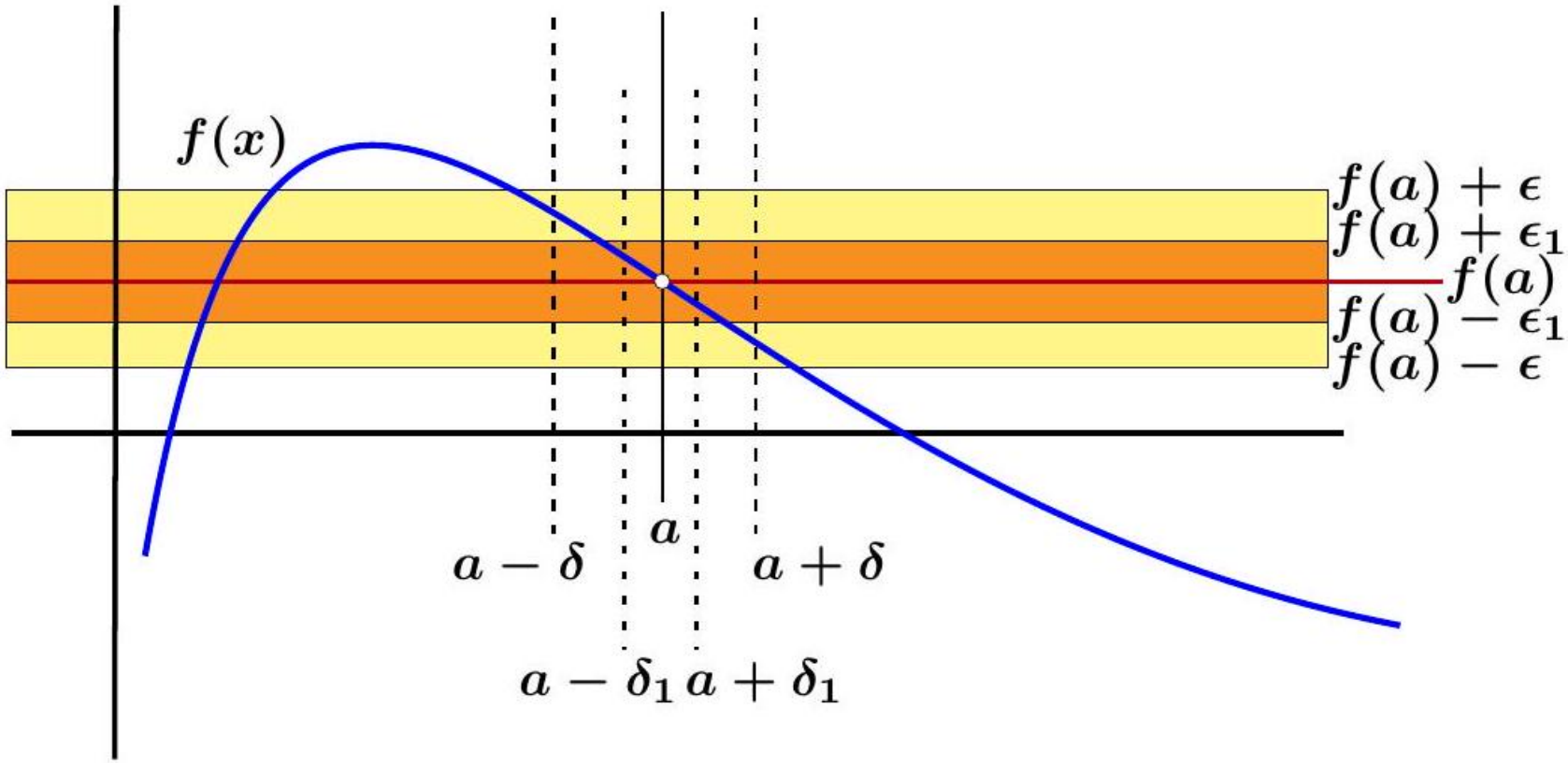
We say that $f(x)$ is continuous at $x = a$ if for every $\epsilon > 0$, there exists a $\delta > 0$ such that if $|x - a| < \delta$, then

$$|f(x) - f(a)| < \epsilon$$

Definition of Continuity



Definition of Continuity



Continuity of Polynomial Functions

Recall: We showed that if

$$p(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

then $\lim_{x \to a} p(x) = p(a)$.

Theorem 2.5.3 (Continuity of Polynomials)

Let

$$p(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

Then $p(x)$ is continuous at each $a \in \mathbb{R}$ with $a > 0$.

Continuity of $\sin(x)$ and $\cos(x)$

Recall: We have seen that

$$\lim_{x\to 0} \sin(x) = 0 = \sin(0) \text{ and } \lim_{x\to 0} \cos(x) = 1 = \cos(0)$$

That is, both $\sin(x)$ and $\cos(x)$ are continuous at $a = 0$.

Theorem 2.5.4 (Continuity of $\sin(x)$ and $\cos(x)$)

Both $\sin(x)$ and $\cos(x)$ are continuous at each $a \in \mathbb{R}$.
Proof: Observe that

$$\begin{aligned}
\lim_{x\to a} \sin(x) &= \lim_{h\to 0} \sin(a+h) \\
&= \lim_{h\to 0} \sin(a)\cos(h) + \sin(h)\cos(a) \\
&= \sin(a)\cdot 1 + 0 \cdot \cos(a) \\
&= \sin(a)
\end{aligned}$$

Continuity of $\sin(x)$ and $\cos(x)$

Proof (continued): Next observe that

$$\begin{aligned}\lim_{x\to a}\cos(x) &= \lim_{h\to 0}\cos(a+h)\\ &= \lim_{h\to 0}\cos(a)\cos(h)-\sin(a)\sin(h)\\ &= \cos(a)\cdot 1-\sin(a)\cdot 0\\ &= \cos(a)\end{aligned}$$

Continuity of e^x

Note:

1. It is actually not an easy task to prove the continuity of the function $f(x) = e^x$.
2. The easiest way to show that e^x is continuous is to realize that it can be defined by a special type of series known as a power series.
3. If e^x is continuous at $x = 0$, then it is continuous everywhere.

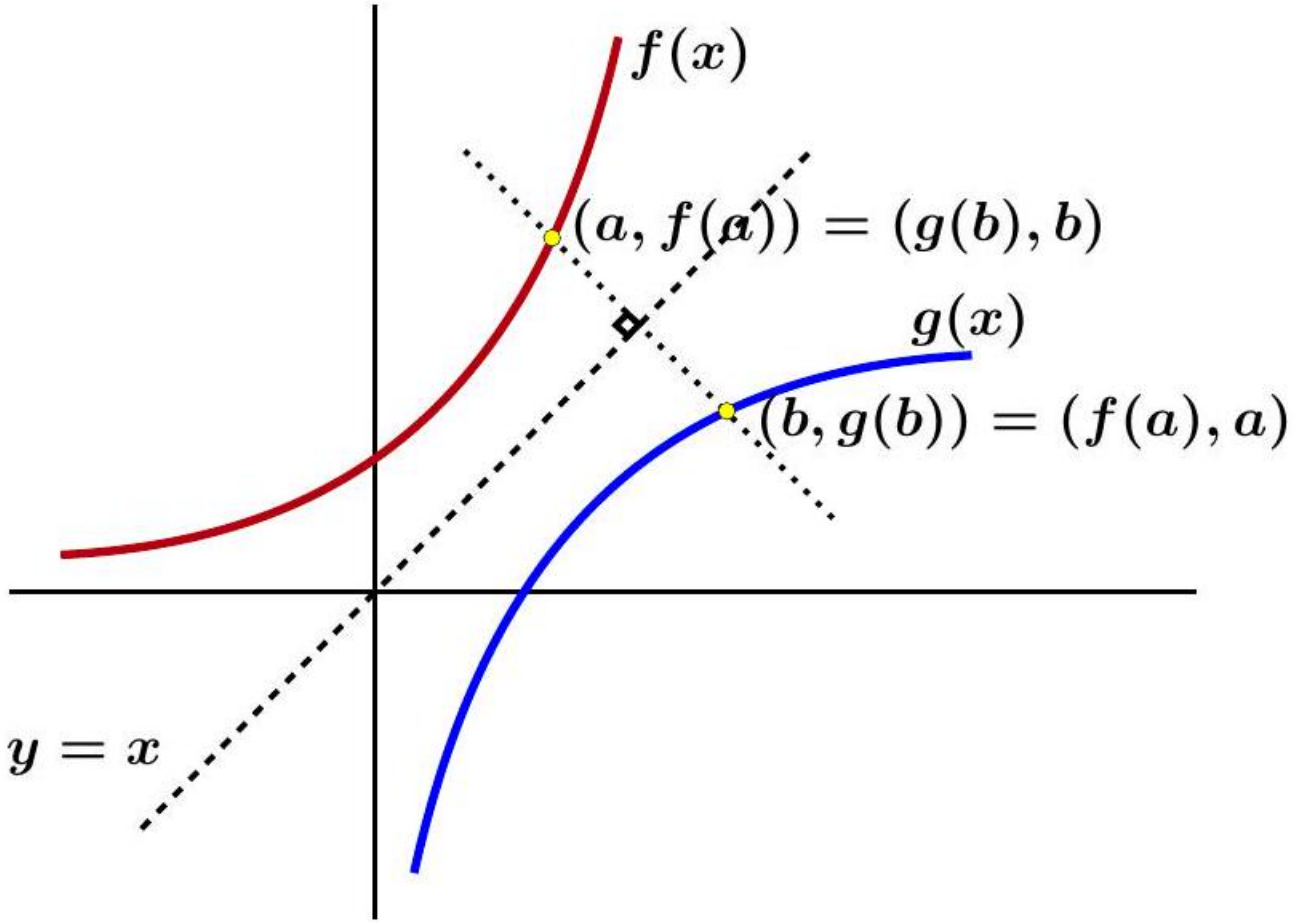
First observe that if e^x is continuous at $x = 0$, then

$$1 = e^0 = \lim_{h \to 0} e^h$$

Hence

$$\begin{aligned} \lim_{x \to a} e^x &= \lim_{h \to 0} e^{a+h} \\ &= \lim_{h \to 0} e^a e^h \\ &= e^a \lim_{h \to 0} e^h = e^a \end{aligned}$$

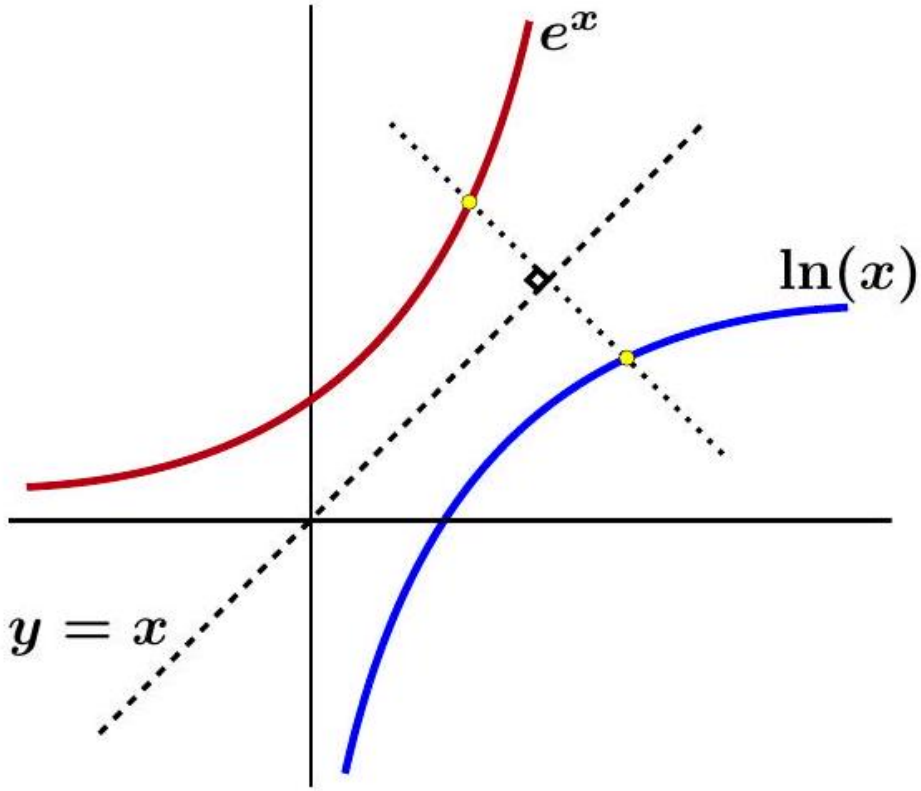
Continuity of Inverses



Theorem 2.5.5 (Continuity of Inverses)

Assume that $f(x)$ is invertible with inverse $g(y)$. If $f(a) = b$ and if $f(x)$ is continuous at $x = a$, then $g(y)$ is continuous at $y = b = f(a)$.

Continuity of e^x and $\ln(x)$



Theorem 2.5.6 (Continuity of e^x and $\ln(x)$)

We have

1. e^x *is continuous at each* $a \in \mathbb{R}$.
2. $\ln(x)$ *is continuous for each* $a \in \mathbb{R}$ *with* $a > 0$.

Arithmetic Rules for Continuity

Theorem 2.5.7 (Arithmetic Rules for Continuity)

If f and g are both continuous at $x = a$, then we have the following:

1. $(cf)(x) = c \cdot f(x)$ *is continuous at* $x = a$ *for all* $c \in \mathbb{R}$.
2. $(f + g)(x) = f(x) + g(x)$ *is continuous at* $x = a$.
3. $(fg)(x) = f(x)g(x)$ *is continuous at* $x = a$.
4. *If* $g(a) \neq 0$, *then* $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$ *is continuous at* $x = a$.

Polynomials and Rational Functions

Remark

1. Let $P(x)$ be any polynomial. Then $P(x)$ is continuous at any $a \in \mathbb{R}$.
2. Let $f(x) = \frac{P(x)}{Q(x)}$ be a rational function. Then $f(x)$ is continuous at $x = a$ if and only if $Q(a) \neq 0$.
3. If a rational function $f(x) = \frac{P(x)}{Q(x)}$ is not continuous at $x = a$, then either $x = a$ is a removable discontinuity or $x = a$ is a vertical asymptote.

Continuity of Composite Functions

Theorem 2.5.8 (Continuity of Composite Functions)

If $f(x)$ is continuous at $x = a$ and $g(y)$ is continuous at $y = f(a)$, then $h(x) = g \circ f(x)$ is continuous at $x = a$.

Proof: Let $x_n \to a$. Then $f(x_n) \to f(a)$ since $f(x)$ is continuous at $x = a$. Since $g(y)$ is continuous at $f(a)$,

$$h(x_n) = g \circ f(x_n) = g(f(x_n)) \to g(f(a)) = g \circ f(a) = h(a)$$

Therefore, $h(x)$ is continuous at $x = a$ by the Sequential Characterization of Continuity.

Example 2.5.9

The function

$$f(x) = e^{x^2 \sin(x)}$$

is continuous on $\mathbb{R}$.

Section 6

Continuity on an Interval

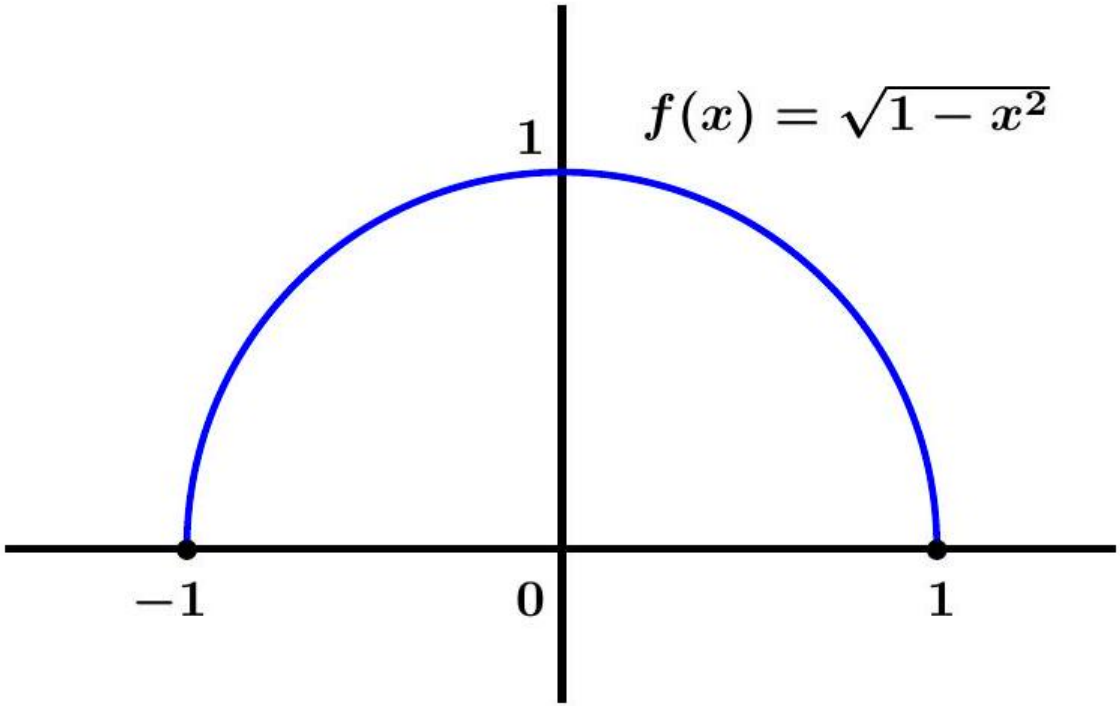
Continuity on a Closed Interval

Definition 2.6.1 (Continuity on a Closed interval)

We say that $f(x)$ is continuous on the closed interval $[a, b]$ if:

1. It is continuous on (a, b). That is $f(x)$ is continuous at every $x \in (a, b)$ in the usual sense.
2. $\lim_{x \to a^+} f(x) = f(a)$.
3. $\lim_{x \to b^-} f(x) = f(b)$.

Continuity on a Closed Interval



Example 2.6.2

Let $f(x) = \sqrt{1 - x^2}$.
We have that

$$f(-1) = \lim_{x \to -1^+} \sqrt{1 - x^2} = 0 = \lim_{x \to 1^-} \sqrt{1 - x^2} = f(1)$$

so $f(x)$ is continuous on $[-1, 1]$.

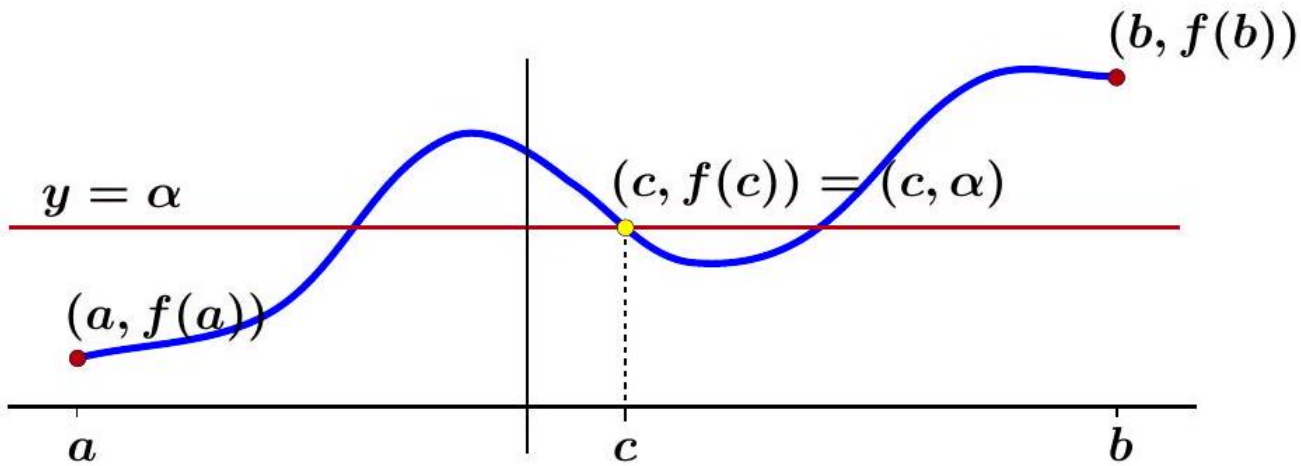
Intermediate Value Theorem (IVT)

Theorem 2.6.3 (The Intermediate Value Theorem (IVT))

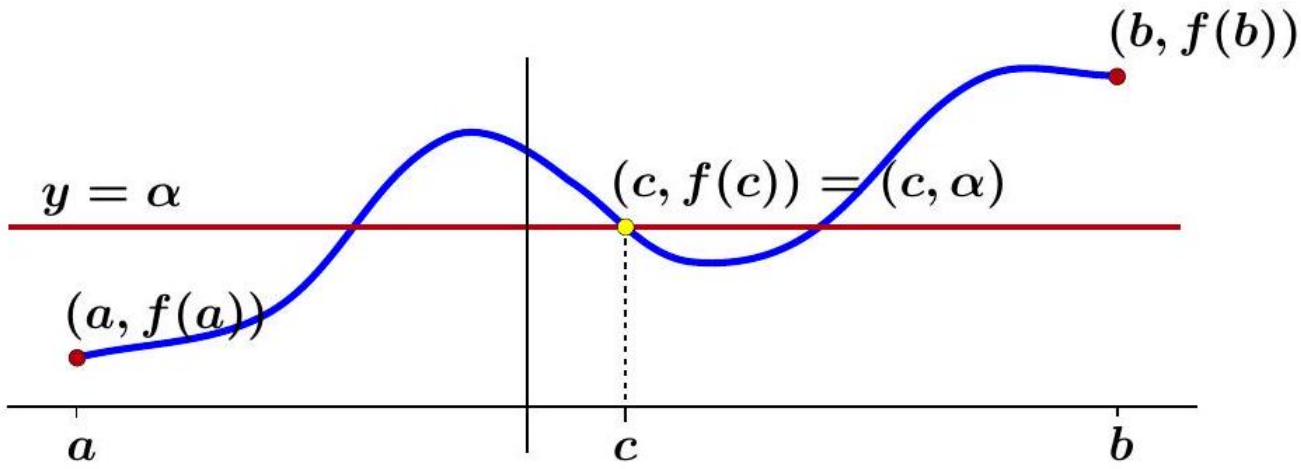
Assume that $f(x)$ *is continuous on the closed interval* $[a, b]$*, and either*

$$f(a) < \alpha < f(b) \text{ or } f(a) > \alpha > f(b)$$

Then there exists a $c \in (a, b)$ *such that* $f(c) = \alpha$.

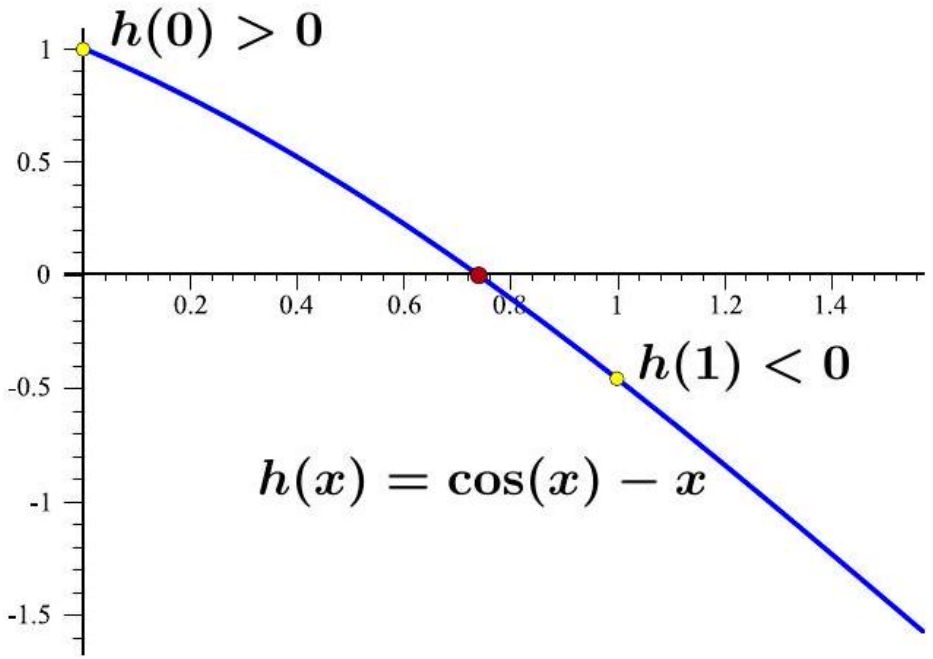


Intermediate Value Theorem (IVT)



Note: The IVT appears to be very obvious but the proof that it is true is actually quite complicated and is beyond the scope of this course.

Example



Example 2.6.4

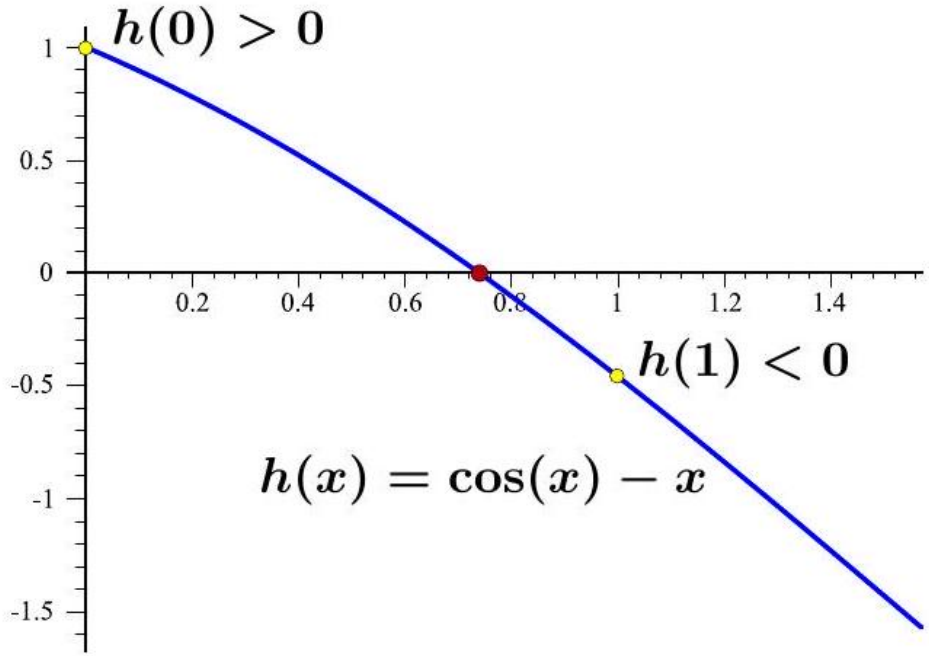
Show that there exists a $c \in (0, 1)$ such that

$$\cos(c) = c$$

or equivalently that there exists a $c \in (0, 1)$ with $h(c) = 0$ where

$$h(x) = \cos(x) - x$$

Example (continued)



Solution: $h(x) = \cos(x) - x$ is continuous on the closed interval $[0, 1]$ with

$$h(0) = \cos(0) - 0 = 1 > 0$$

and

$$h(1) = \cos(1) - 1 < 0$$

By the IVT, we can conclude that there is a $0 < c < 1$ such that $h(c) = 0$ or equivalently that $\cos(c) = c$.

The Intermediate Value Theorem

Theorem 2.6.5 (Intermediate Value Theorem (IVT))

Assume that $f(x)$ is continuous on $[a, b]$ and that $f(a) \cdot f(b) < 0$. Then there exists $c \in (a, b)$ such that

$$f(c) = 0.$$

Note: We have $f(a) \cdot f(b) < 0$ if and only if $f(a) < 0 < f(b)$ or $f(b) < 0 < f(a)$.

Question: How can we estimate c within a given error?

Extreme Value Theorem

Theorem 2.6.6 (The Extreme Value Theorem (EVT))

Suppose that $f(x)$ *is continuous on* $[a, b]$. *There exists* c_1 *and* $c_2 \in [a, b]$ *such that*

$$f(c_1) \leq f(x) \leq f(c_2)$$

for all $x \in [a, b]$.

Question: Is continuity important?

Example

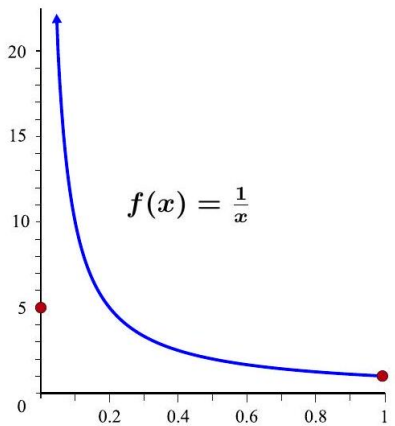
Example 2.6.7

Let

$$f(x) = \begin{cases} \frac{1}{x} & \text{if } 0 < x \leq 1 \\ 5 & \text{if } x = 0 \end{cases}$$

Then $f(x)$ has a global minimum on $[0, 1]$ at $x = 1$, but it has no global maximum on $[0, 1]$.

Note: This example does not contradict the EVT because $f(x)$ is not continuous on $[0, 1]$.



Extreme Value Theorem

Observation: The EVT ensures that a continuous function $f(x)$ defined on a closed interval $[a, b]$ achieves its global maximum and minimum on $[a, b]$, but it does not tell us how to find these values.

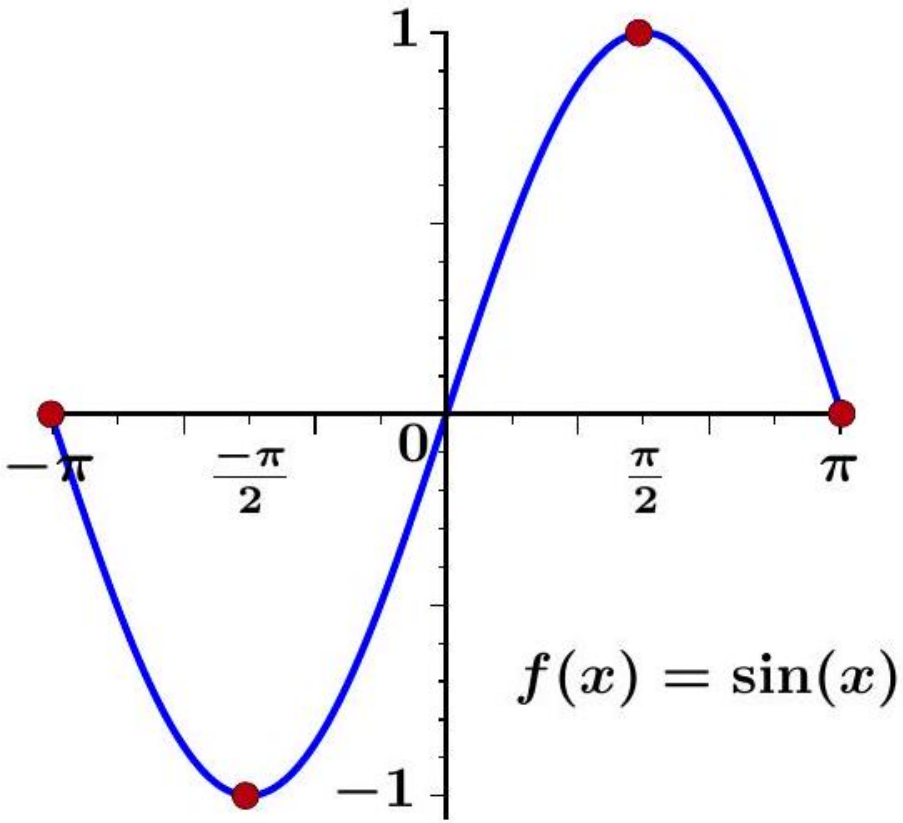
Important Fact: Assume that $f(x)$ has either a global maximum or global minimum at $c \in [a, b]$. Then either

- 1) c is an end point of the interval $[a, b] \Rightarrow c = a$ or $c = b$

or

- 2) c is not an end point of the interval $[a, b] \Rightarrow c \in (a, b)$.

Example



Example 2.6.8

The function $f(x) = \sin(x)$ assumes its maximum and minimum values on $[-\pi, \pi]$ at $x = \frac{\pi}{2}$ and $x = -\frac{\pi}{2}$, respectively.

Extreme Value Theorem

Important Fact:
Continuous functions can behave very differently on closed intervals than they do on open intervals.